BITS Pilani

Embedded System Design

Sangmeshwar Kendre

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BITS Pilani

AEZG512/AELZG512/DMZG512
Embedded System Design
Remote Lab Contact Session-01

Contents

- Instructions
- Remote Lab Demonstration

Instructions to be followed

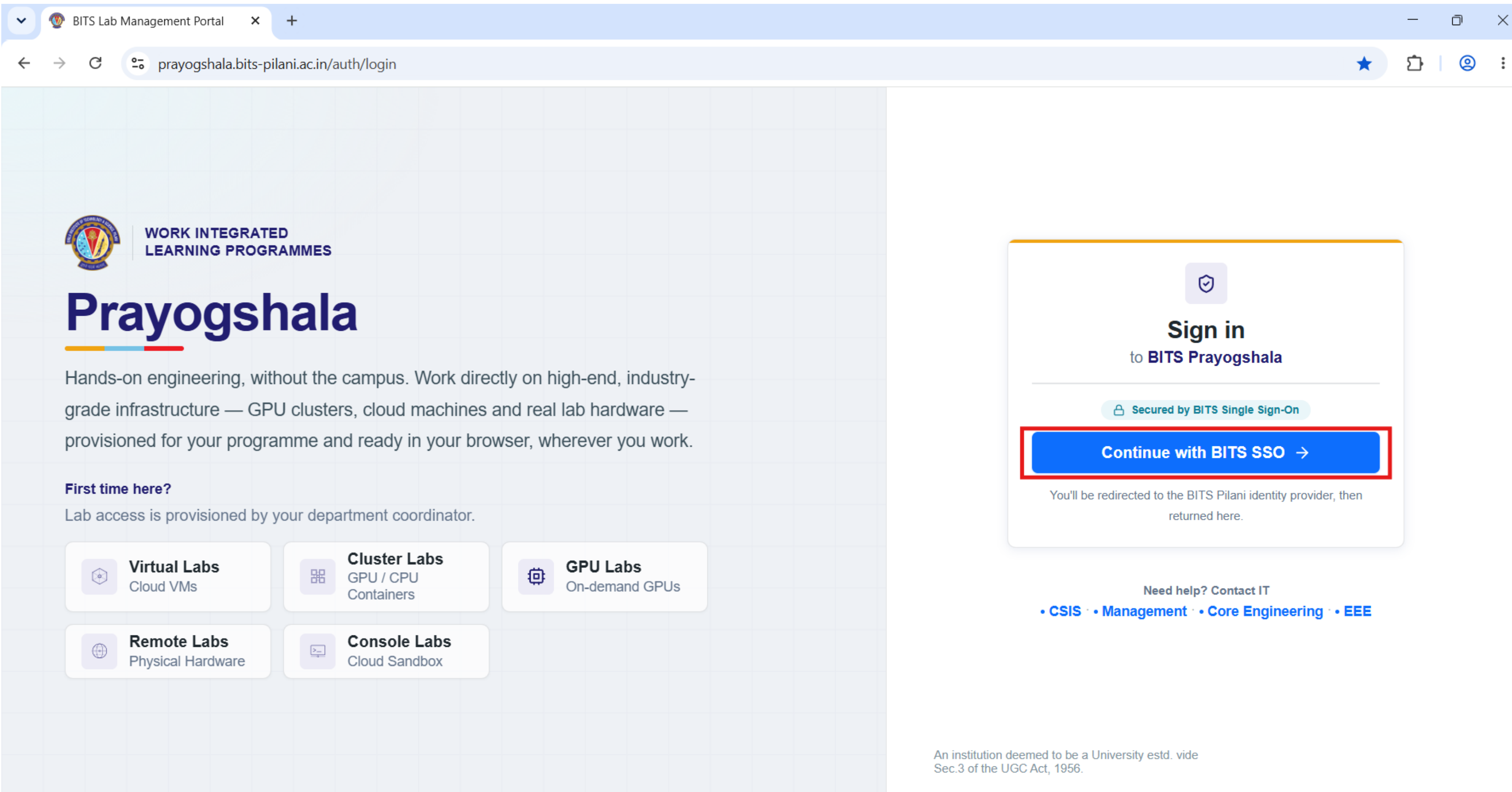
- You MUST have BITS ID and password to access the lab. Use same credentials as that of http://elearn.bits-pilani.ac.in
- Everyone should complete the class recordings before accessing the remote lab.
- Unfortunately, no additional slots will be available for you under any circumstances.
- Ensure the recordings, documents, and remote assignments to be performed are in the elearn portal or under the teams before accessing the lab.
- Only one slot will be available with two hours of duration.
- The cancellation of slots will only be possible before 24 hours, and you can rebook again, which is only possible for one time.

Instructions to be followed

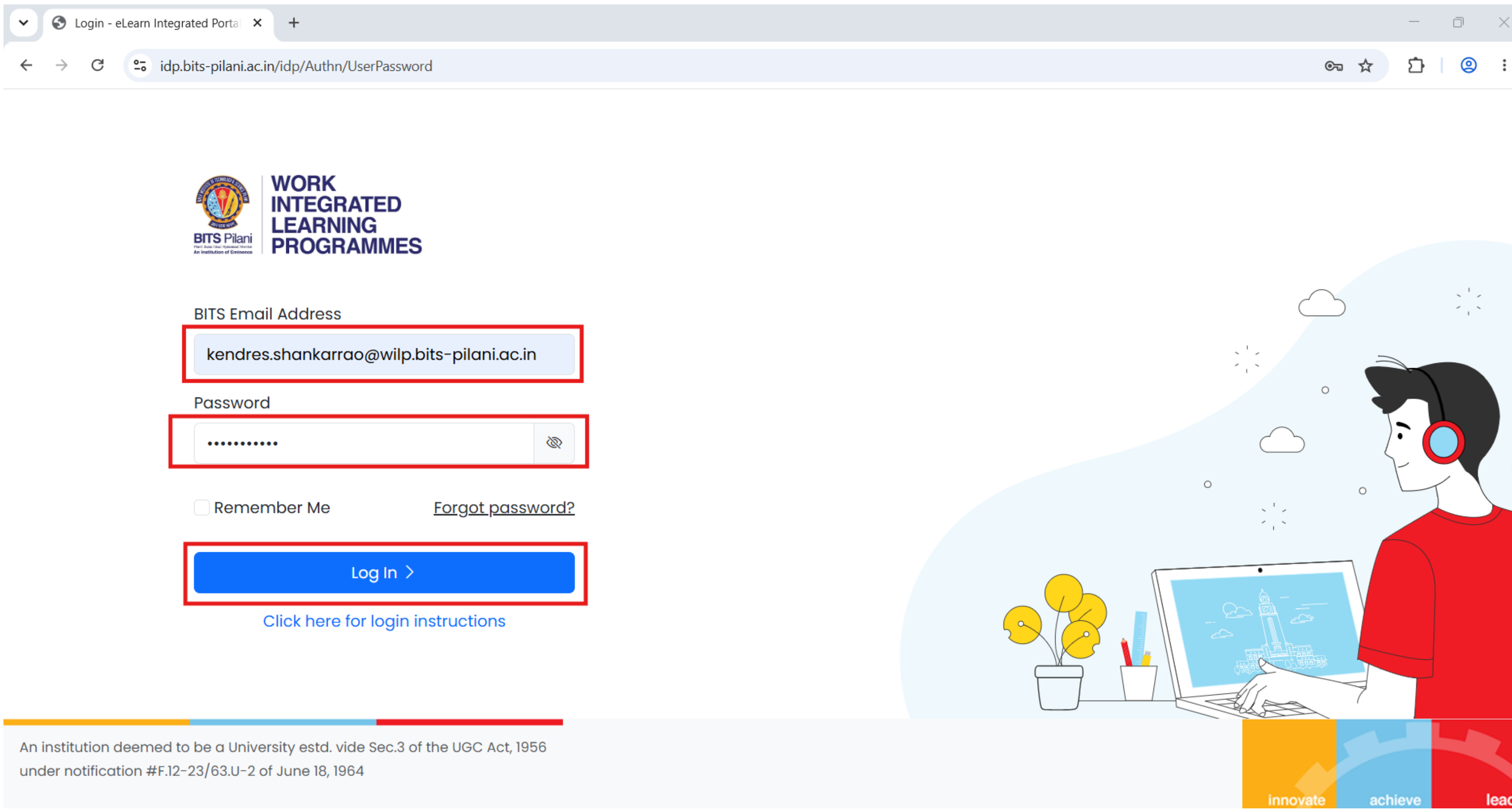
- Verify that RDP is enabled on your computer; if not, follow the steps: When you're ready, select Start > System > Remote Desktop, and turn on Enable Remote Desktop
- Don’t use office laptops as they are equipped with security firewalls.

Remote Lab Booking

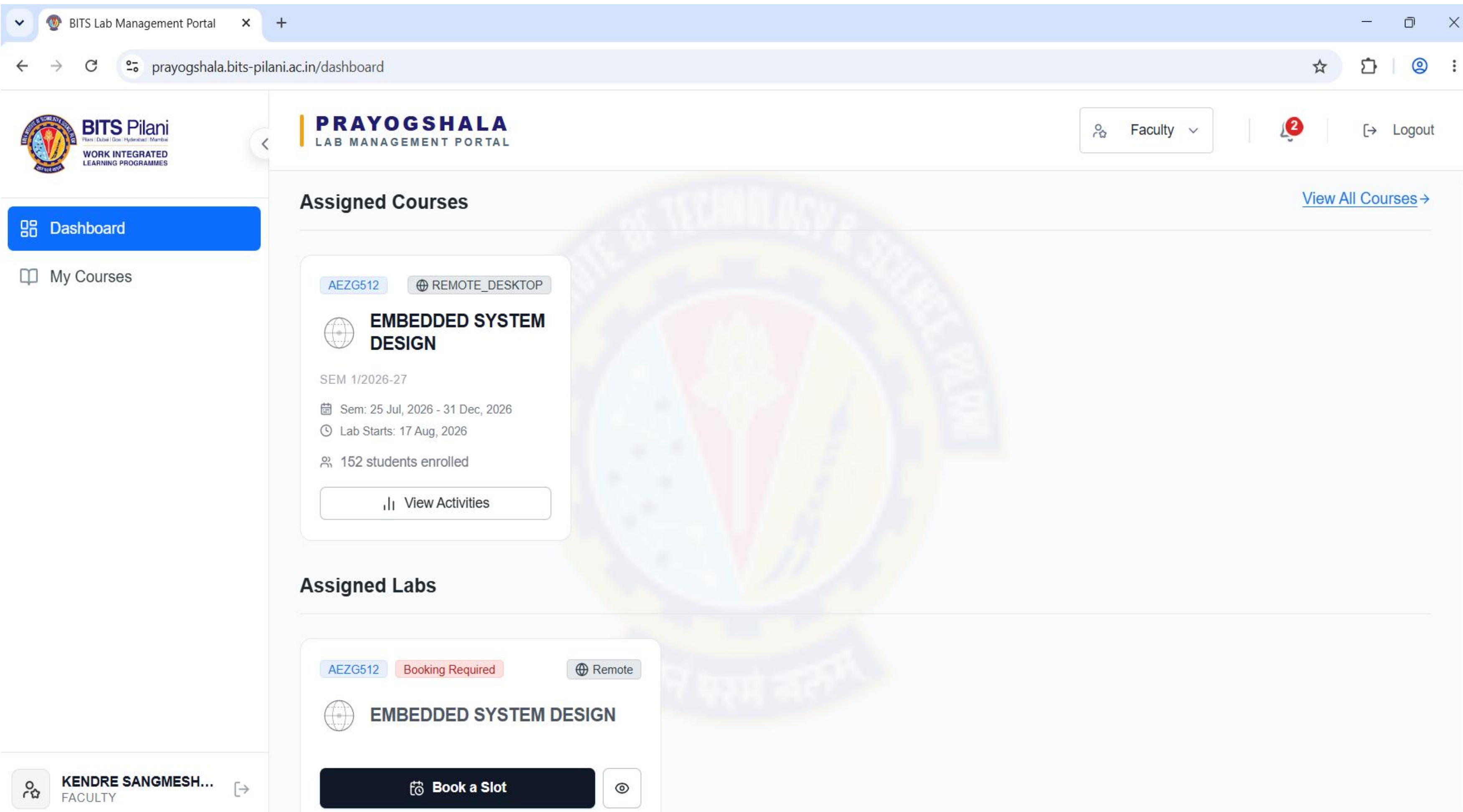
Go to https://prayogshala.bits-pilani.ac.in/auth/login **Click on Continue with BITS SSO**



Enter your BITS ID and password to login (Use same credentials as that of http://elearn.bits-pilani.ac.in). Click on Login

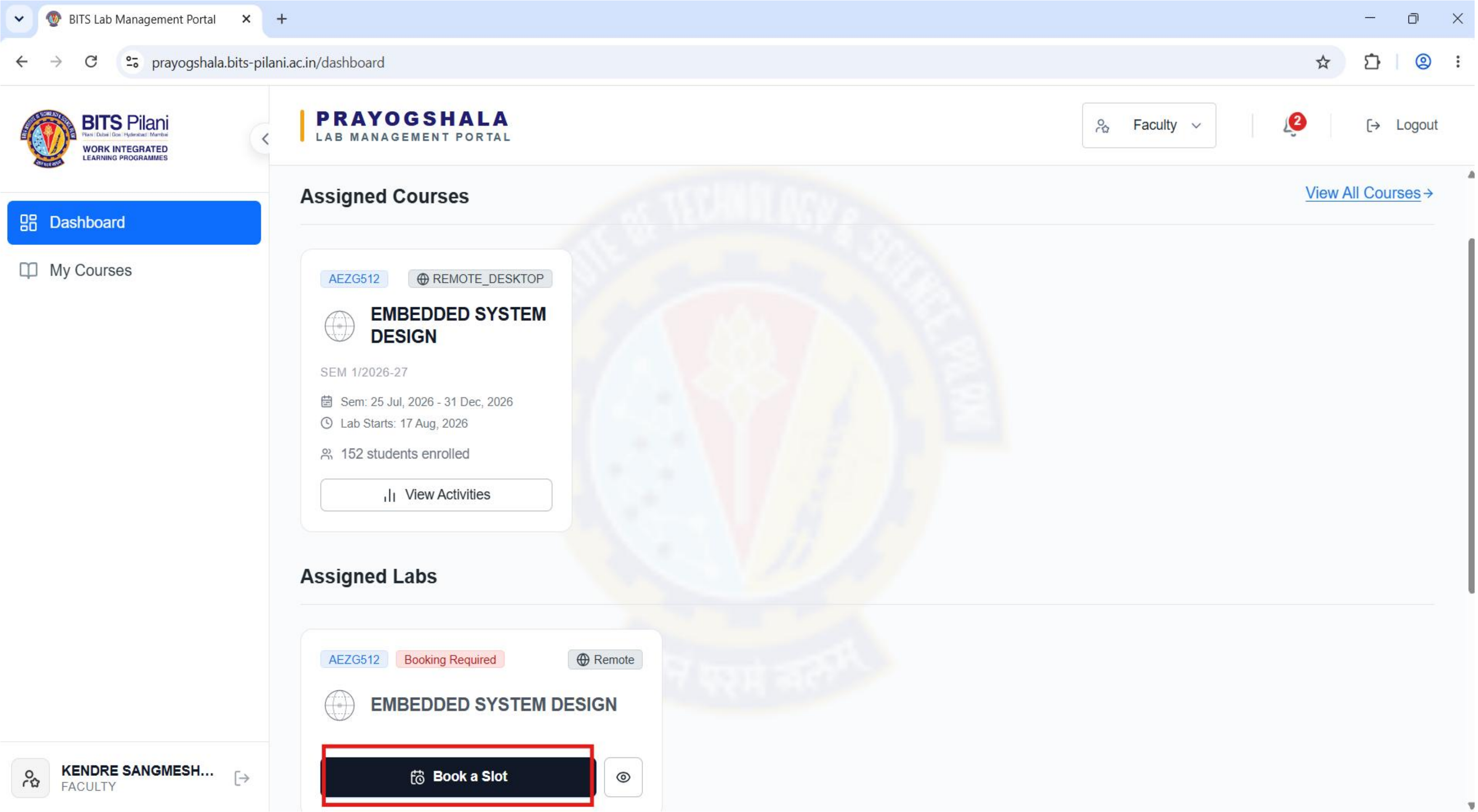


After successful login

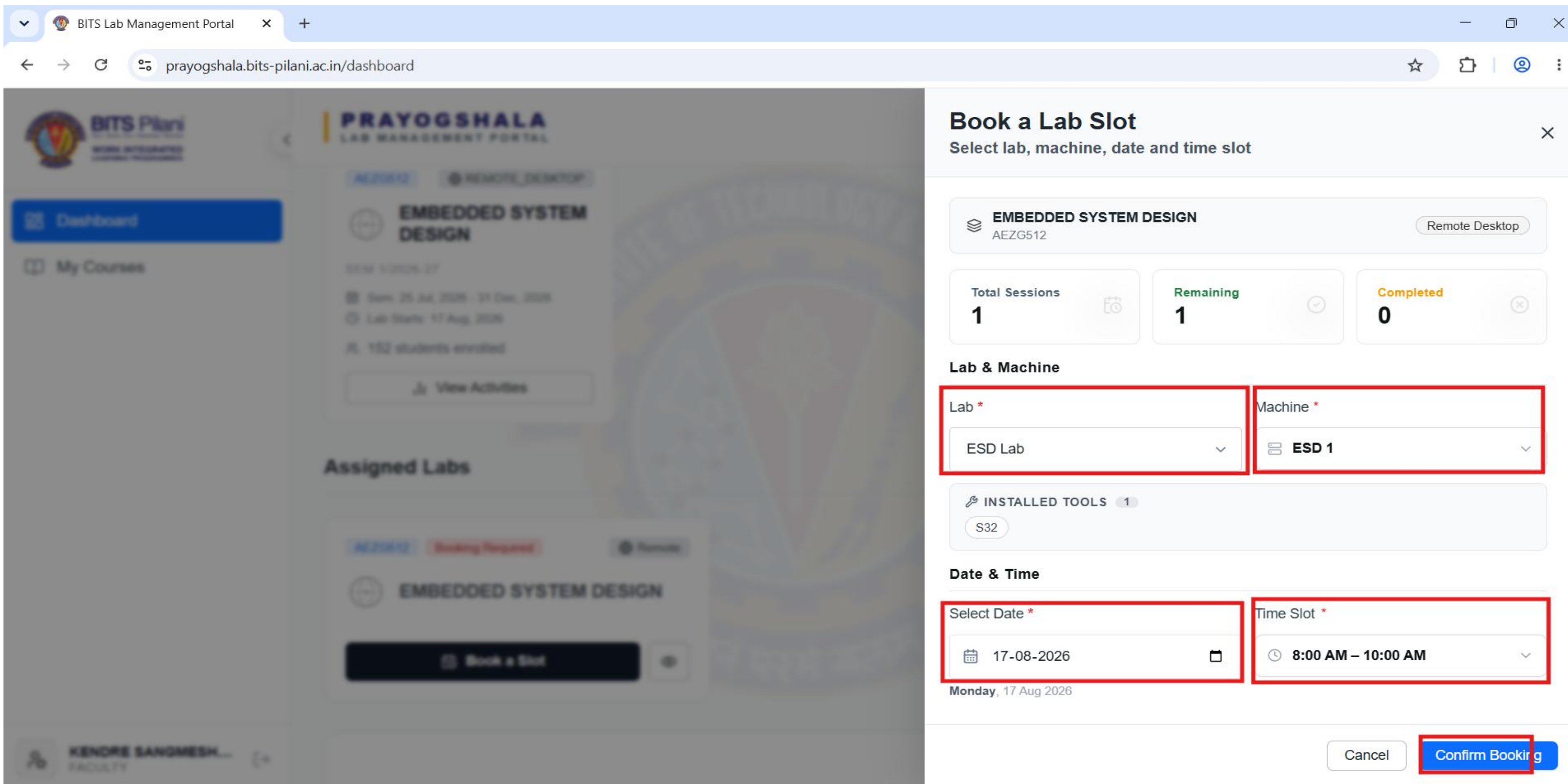


For Remote Lab Booking

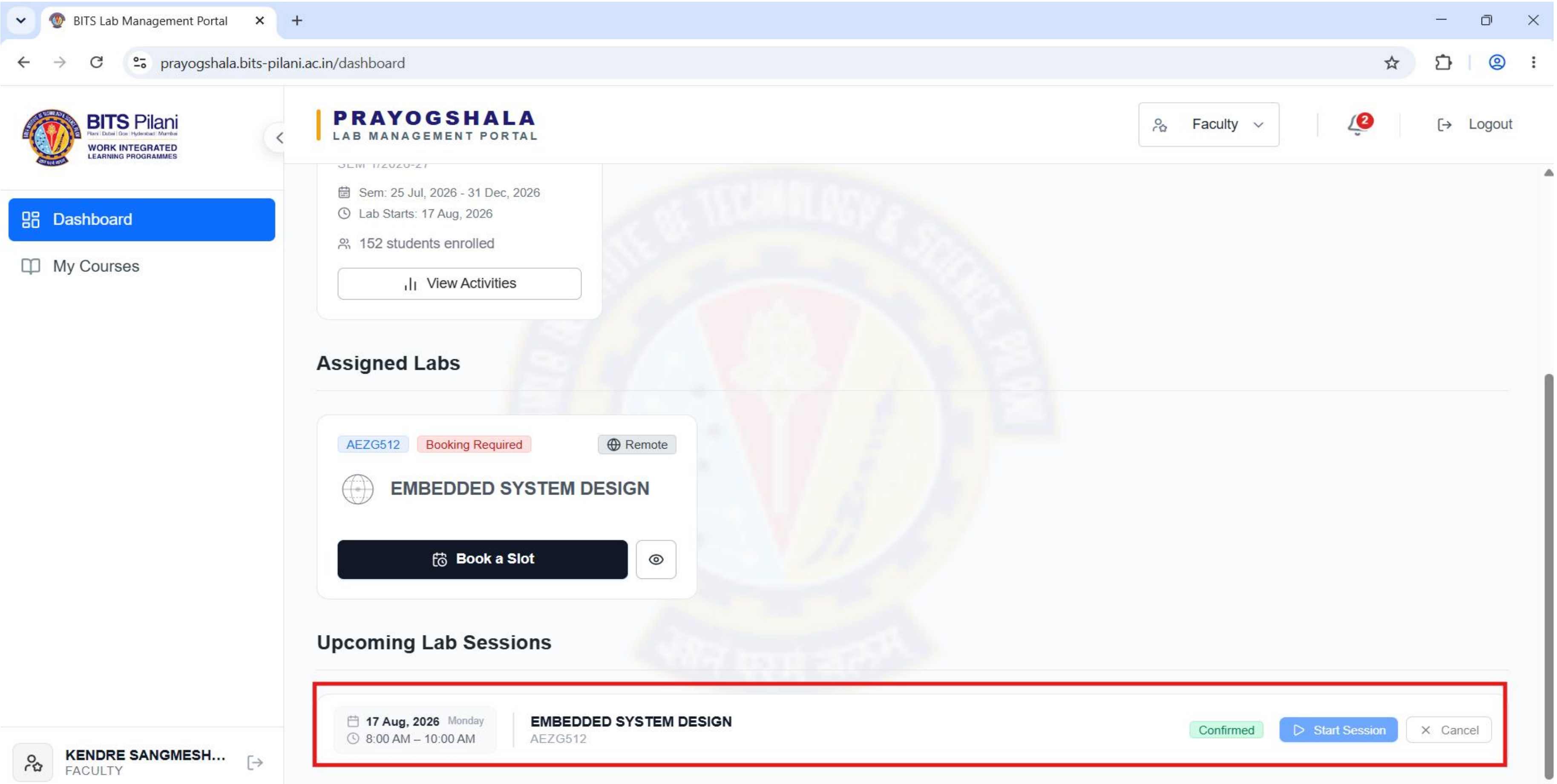
Select the course: EMBEDDED SYSTEM DESIGN and click on Book a slot



Select the Lab: **ESD Lab**, Machine: **ESD1**, Date: **17-08-2026** and Time slot: **8:00 AM to 10:00 AM**. Click on **Confirm Booking**

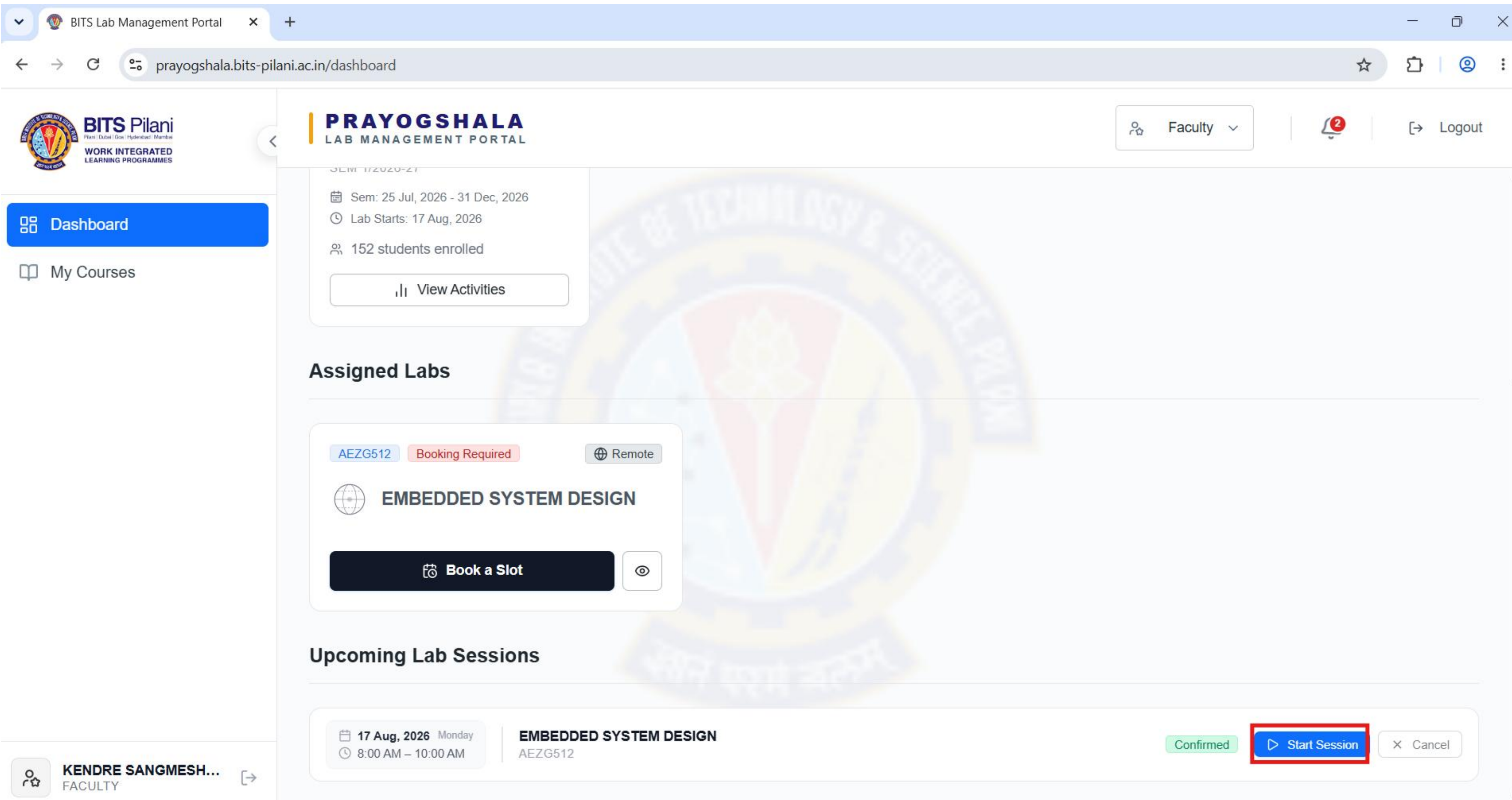


You will see booked lab session. Start Session button will be enabled at the appointed date and time only. Click on it

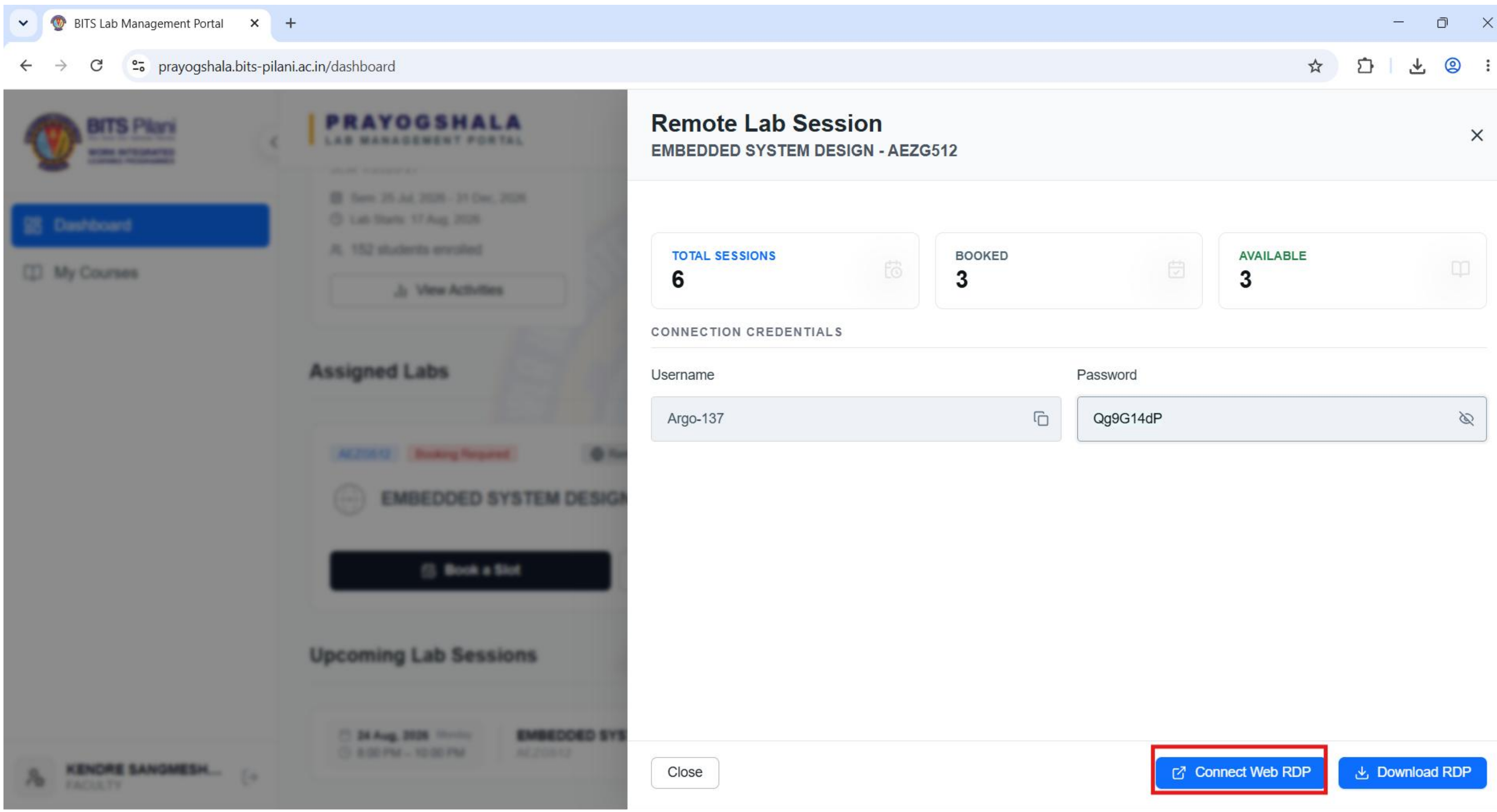


To access remote lab

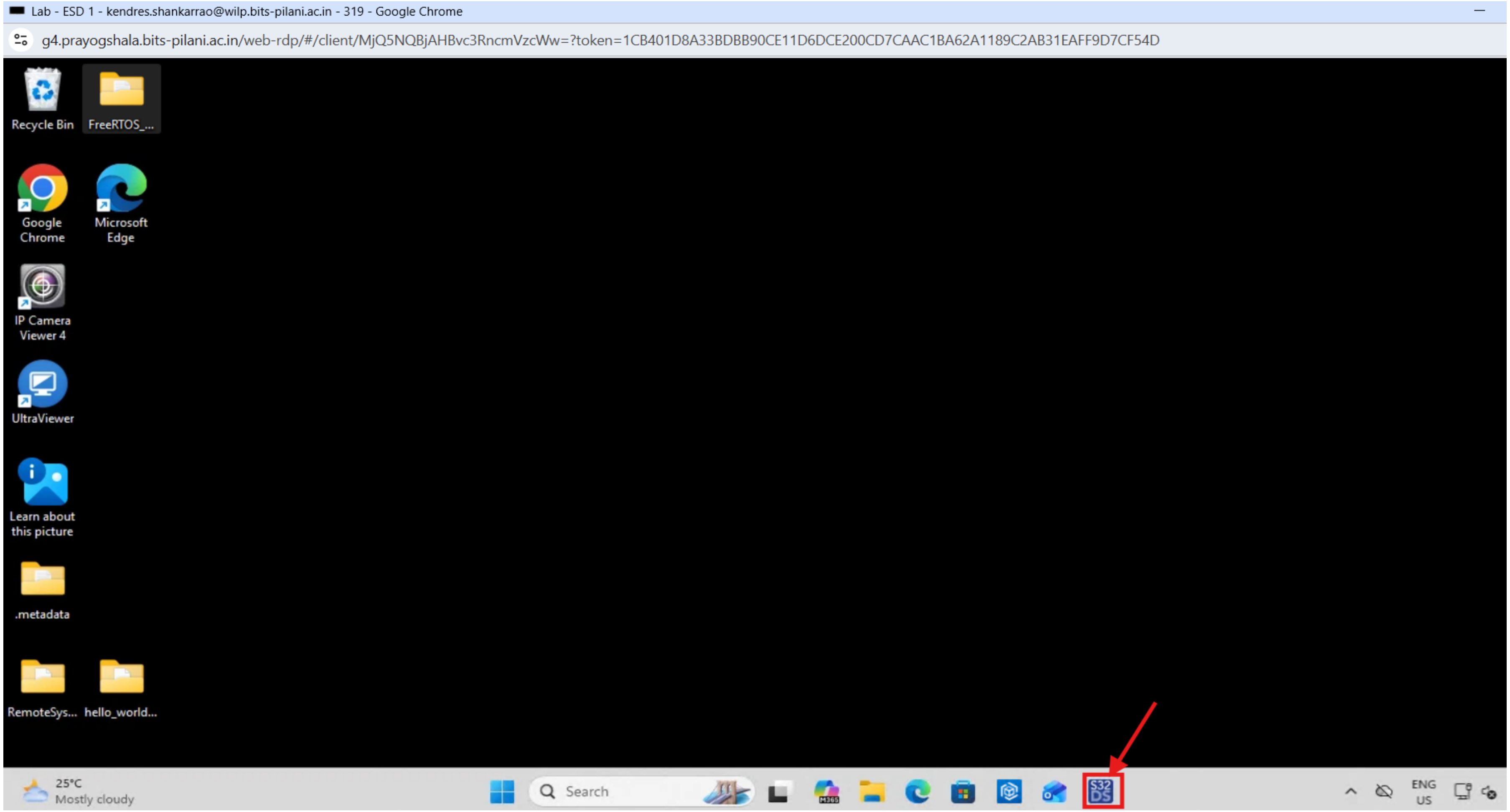
Click on **Start Session**



Option-1: To connect remote through Web RDP PC, Click on **Connect Web RDP**



Then it will be connected to remote lab using browser.



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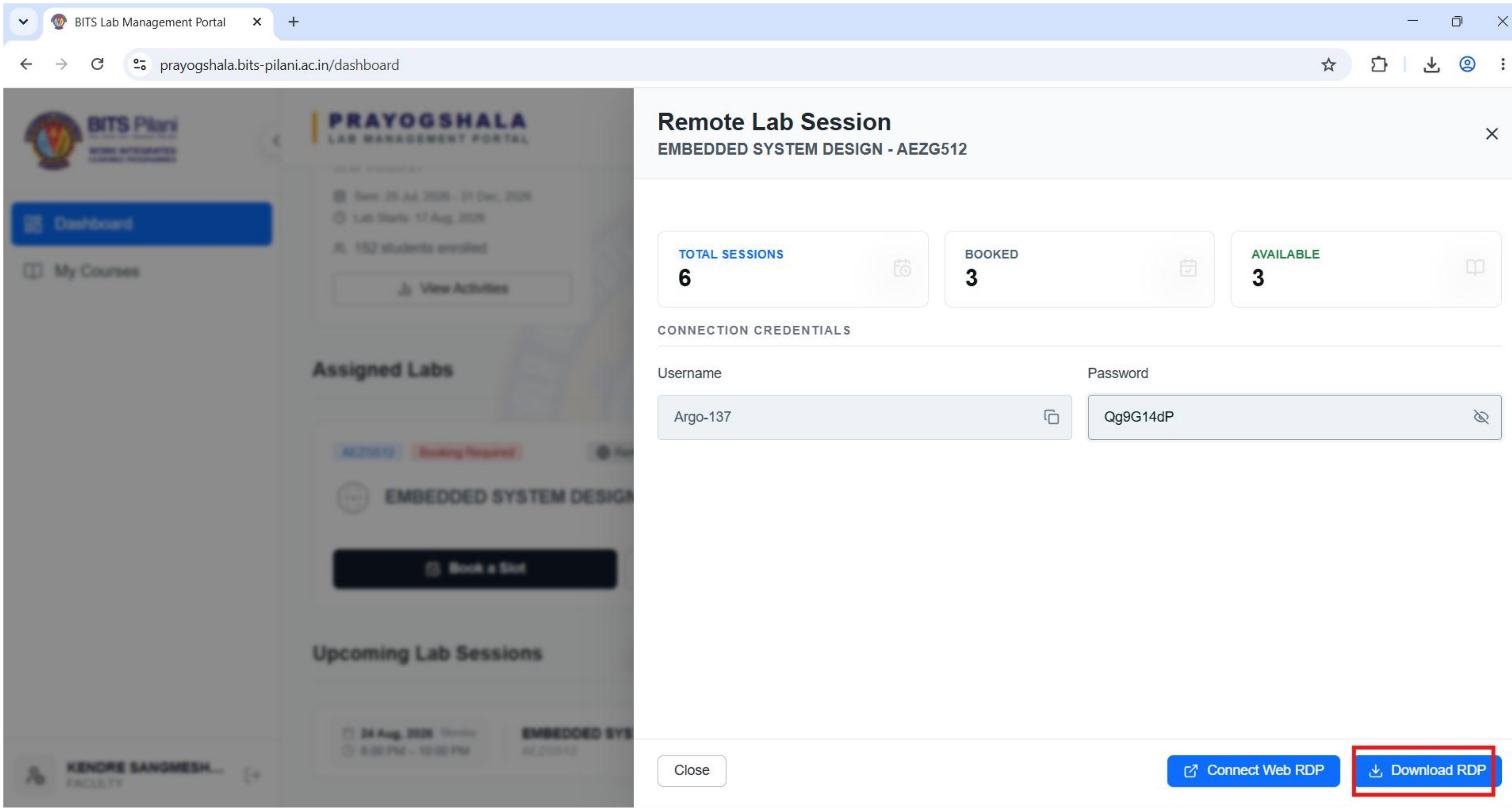
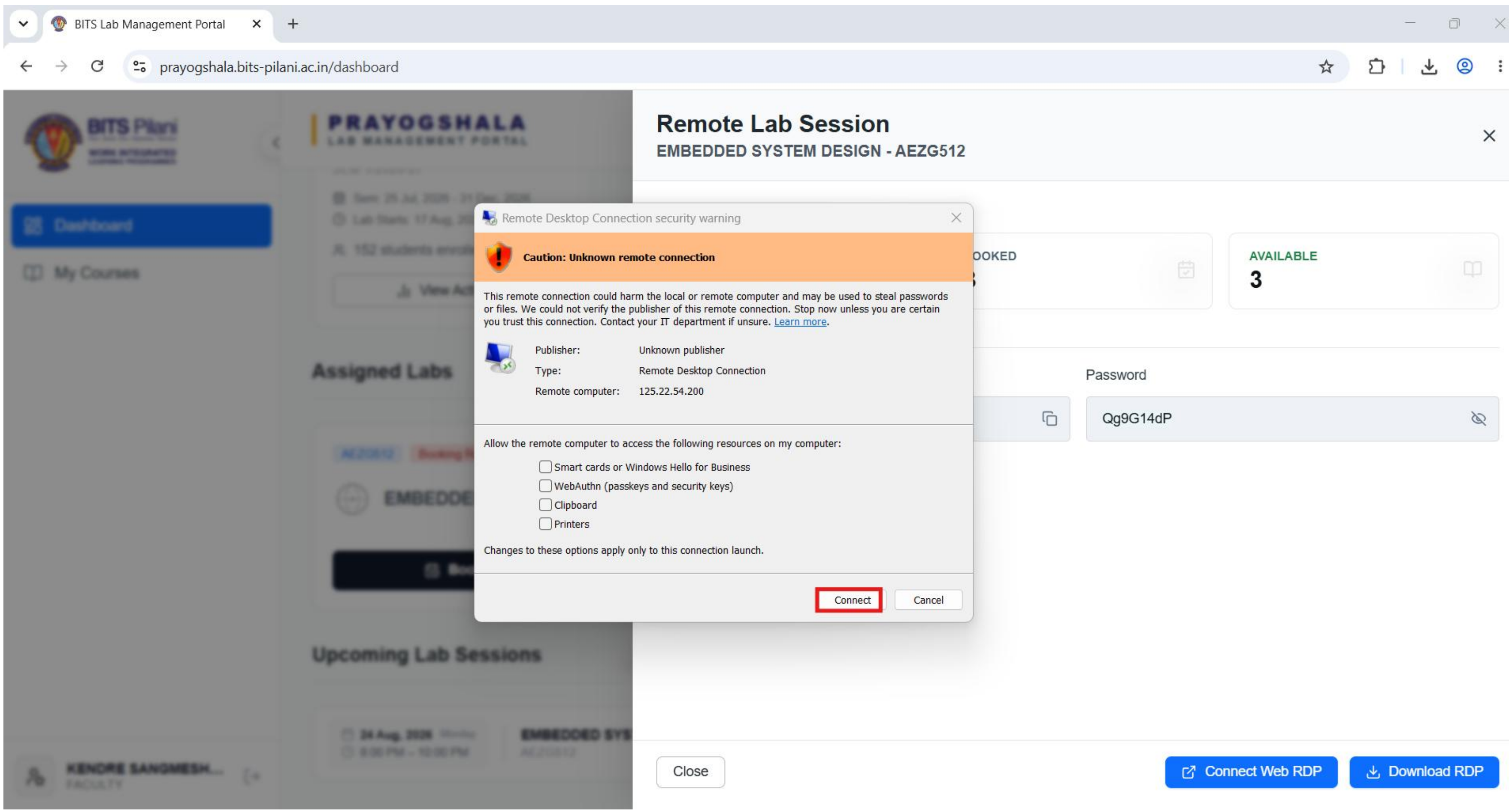
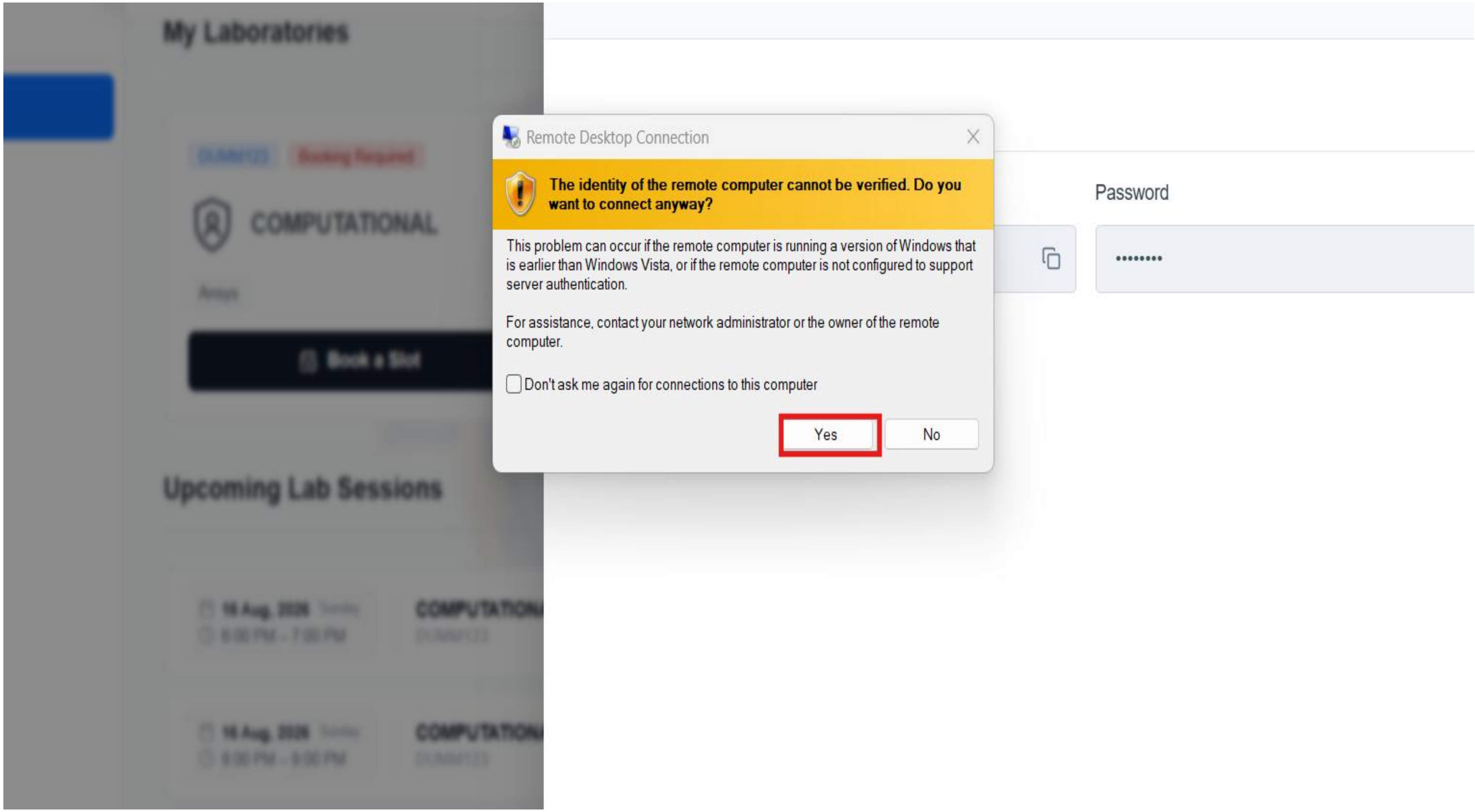
Option-2: Download RDP. Click on **Download RDP**



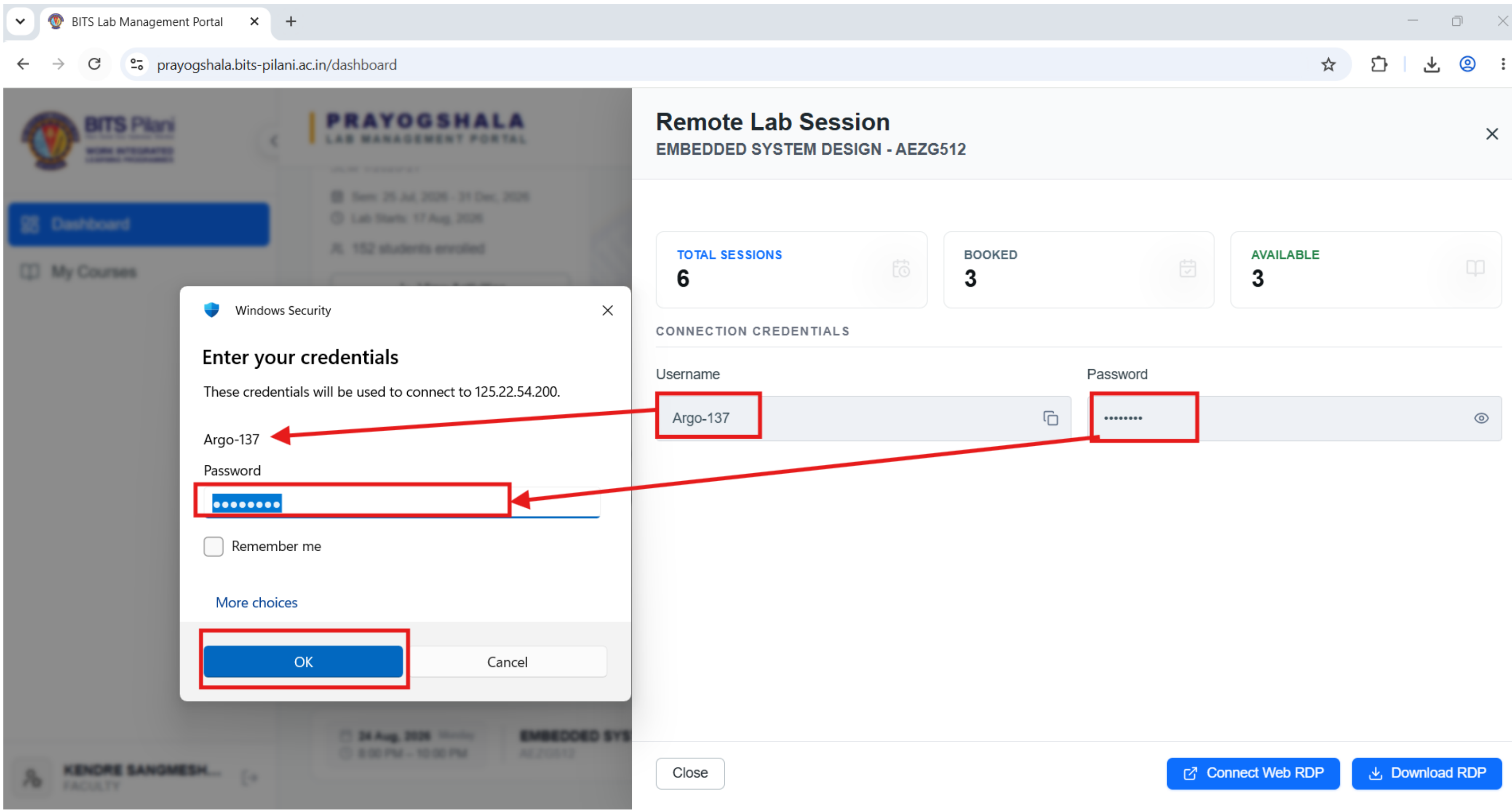
Double Click on downloaded RDP file. Click on Connect



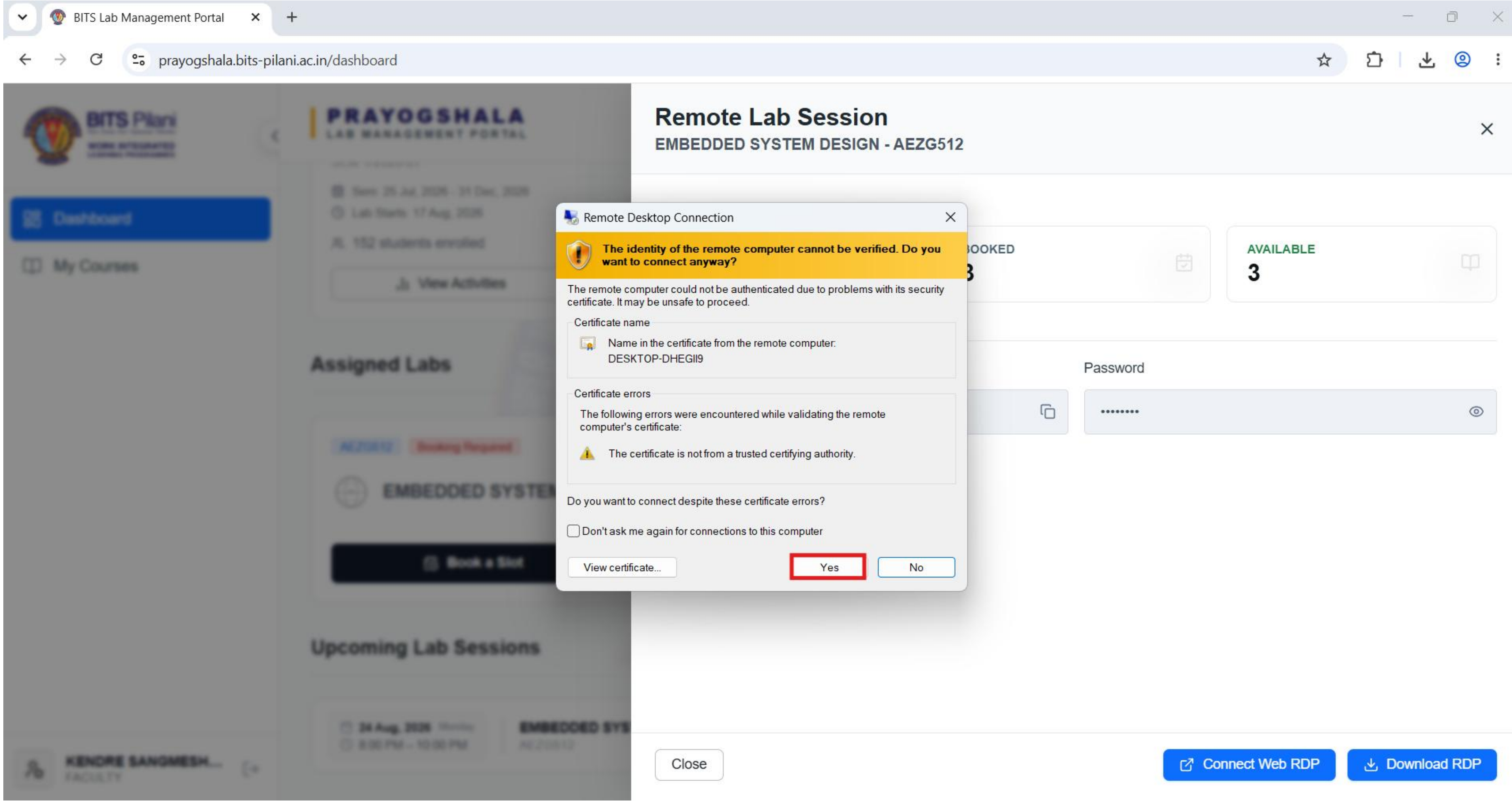
Click Yes


My Laboratories
COMPUTATIONAL
Upcoming Lab Sessions
Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
This problem can occur if the remote computer is running a version of Windows that is earlier than Windows Vista, or if the remote computer is not configured to support server authentication.
For assistance, contact your network administrator or the owner of the remote computer.
Don't ask me again for connections to this computer
Yes
No
Password

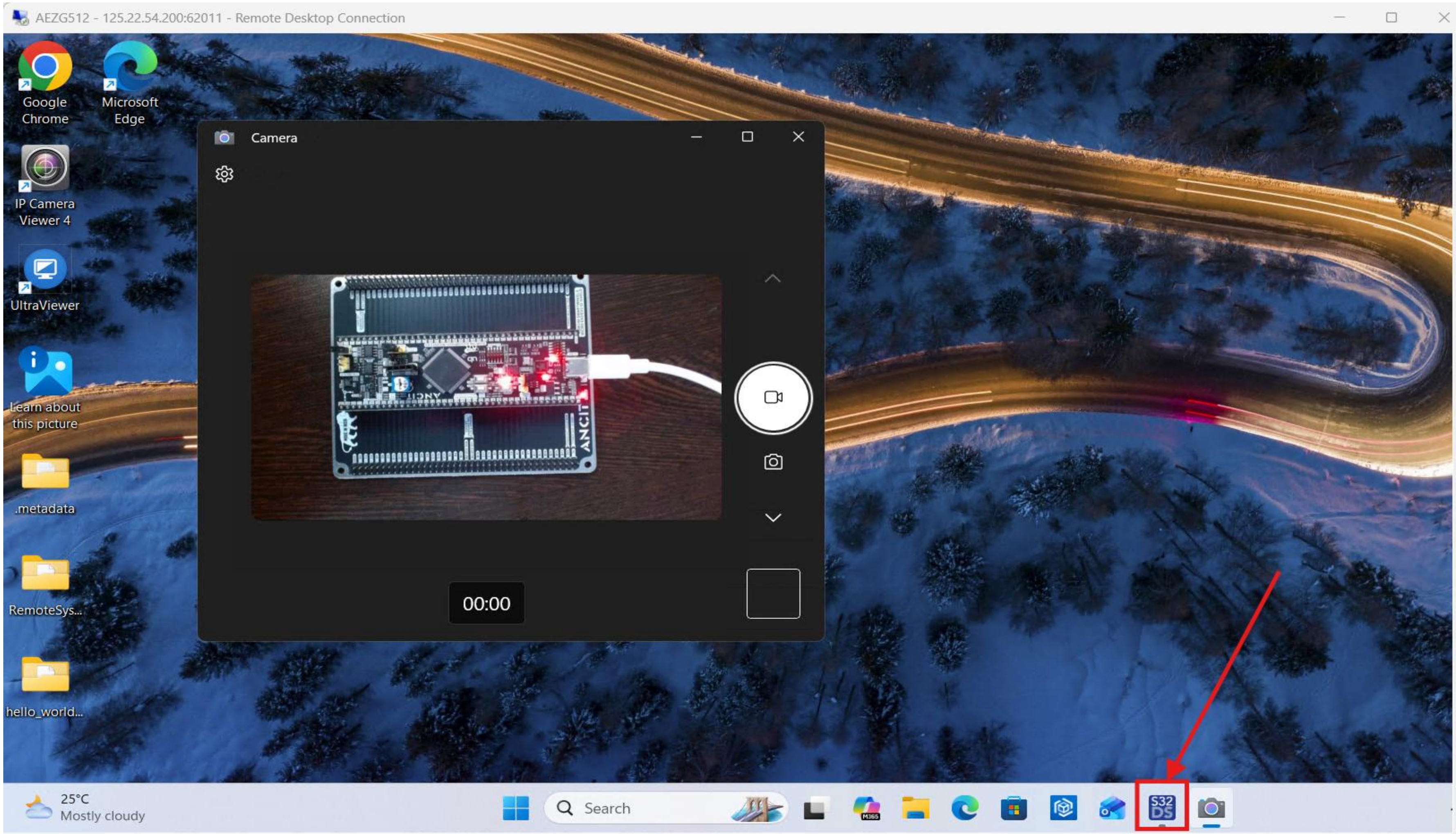
Enter your Credentials given by the portal, password. Then click OK



Click on Yes.

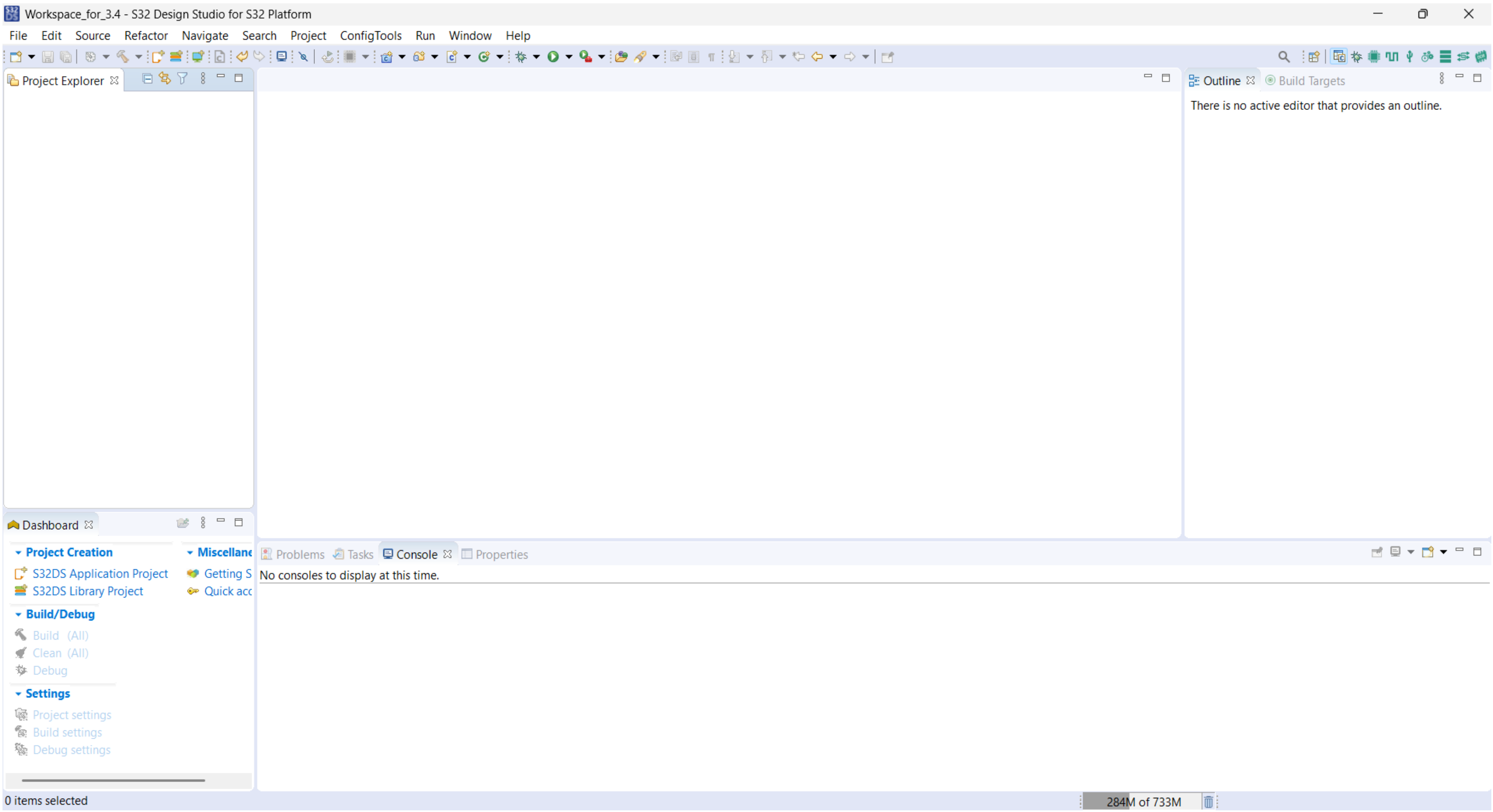

BITS Lab Management Portal
prayogshala.bits-pilani.ac.in/dashboard
Remote Lab Session
EMBEDDED SYSTEM DESIGN - AEZG512
Remote Desktop Connection
The identity of the remote computer cannot be verified. Do you want to connect anyway?
The remote computer could not be authenticated due to problems with its security certificate. It may be unsafe to proceed.
Certificate name
Name in the certificate from the remote computer:
DESKTOP-DHEGII9
Certificate errors
The following errors were encountered while validating the remote computer's certificate:
The certificate is not from a trusted certifying authority.
Do you want to connect despite these certificate errors?
Don't ask me again for connections to this computer
View certificate...
Yes
No
AVAILABLE
3
Password
Close
Connect Web RDP
Download RDP

Remote Desktop opens. Start S32 Design studio and Camera from application menu.

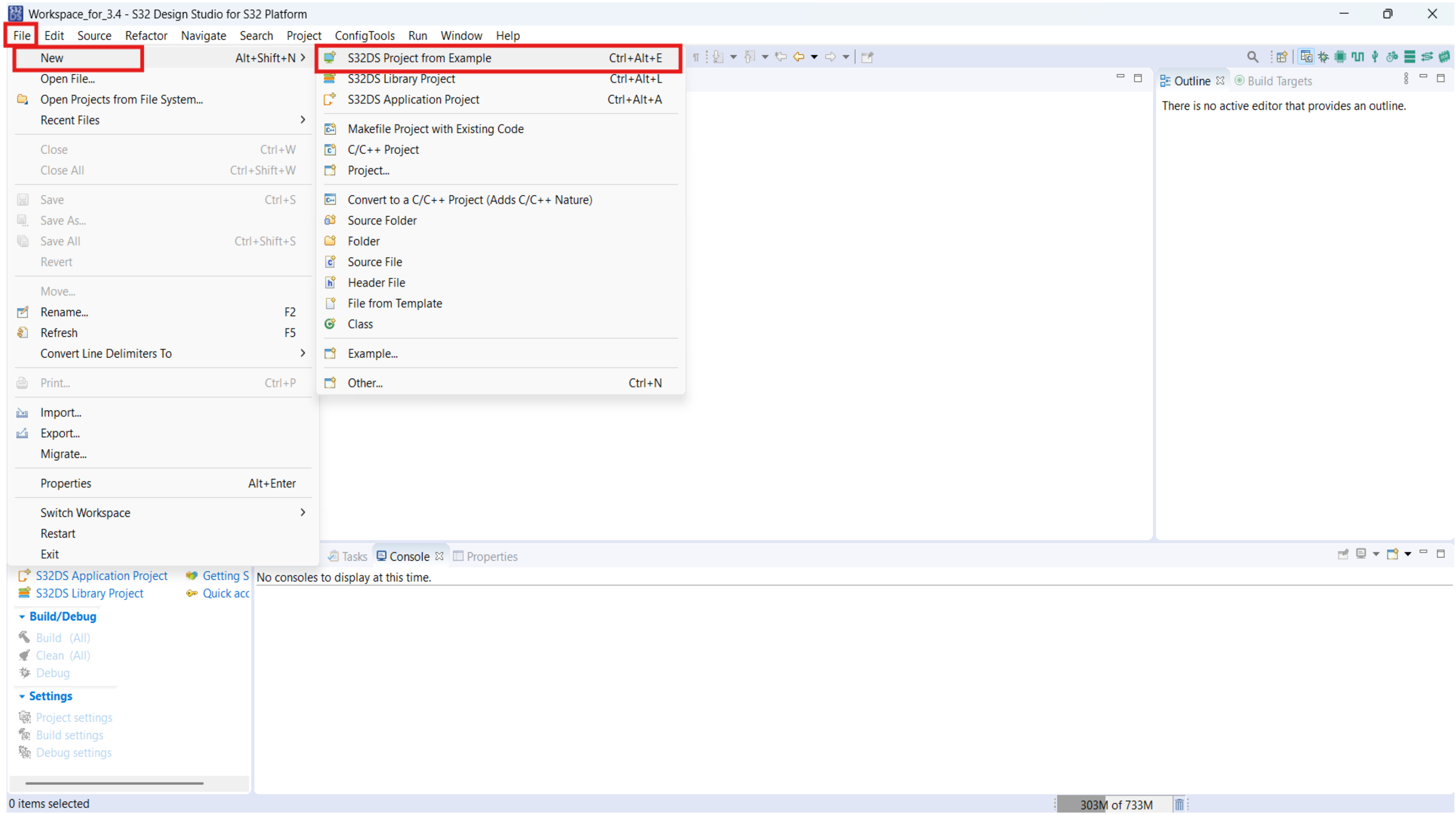


Project Creation in S32 Design Studio

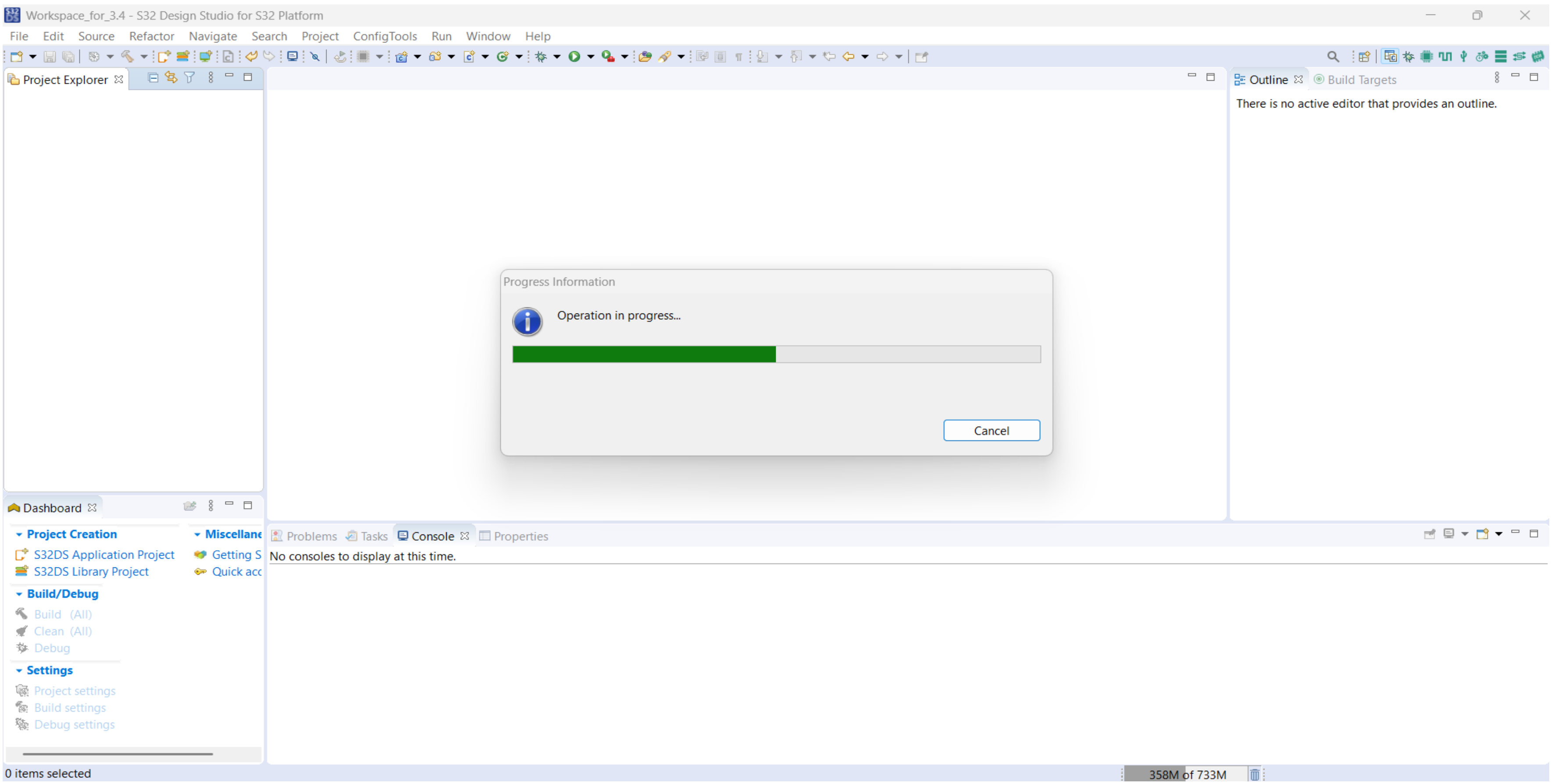
S32 Design Studio for S32 Platform 3.4



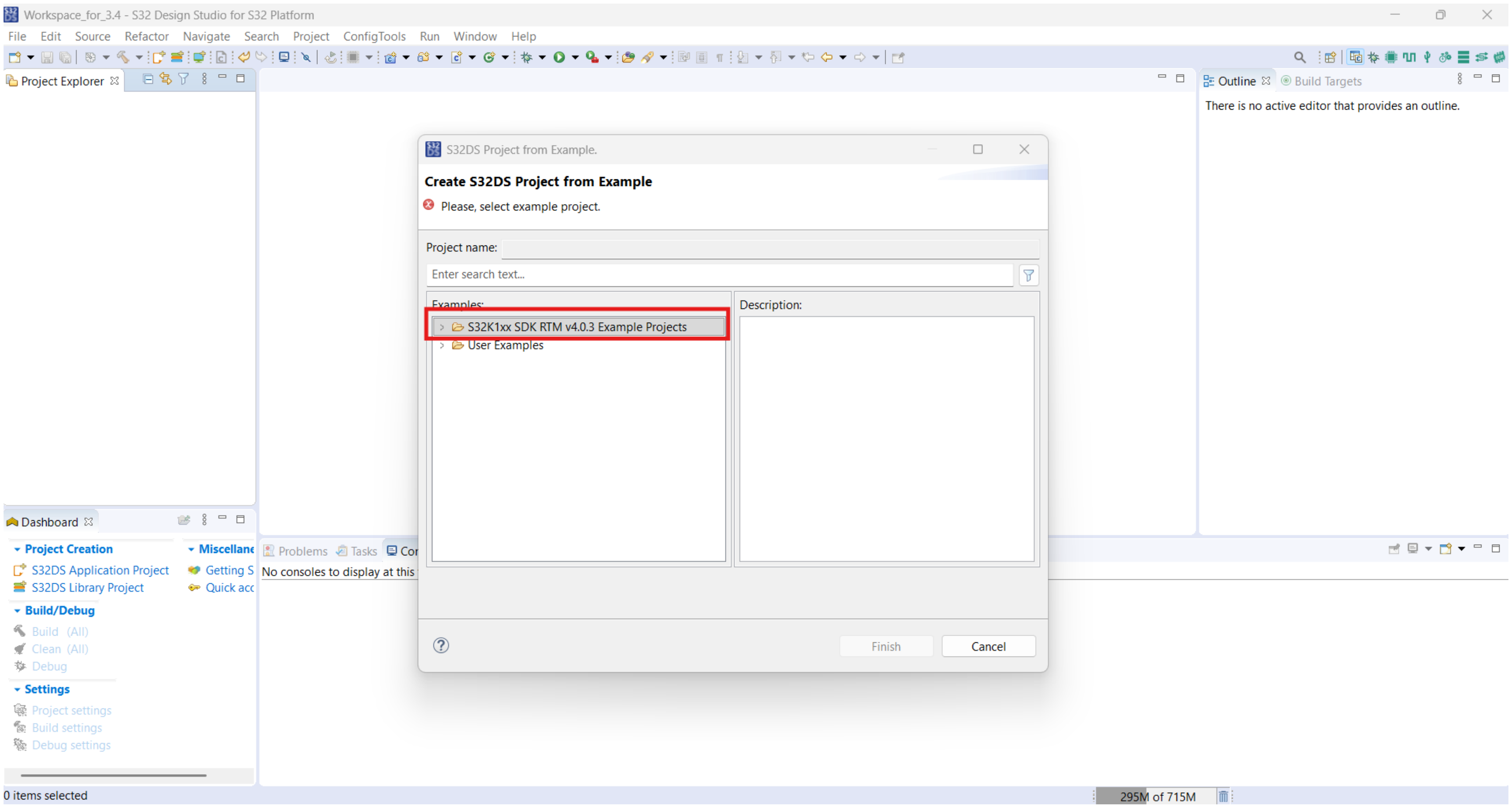
Navigate to File–>New–>S32DS Project from Example



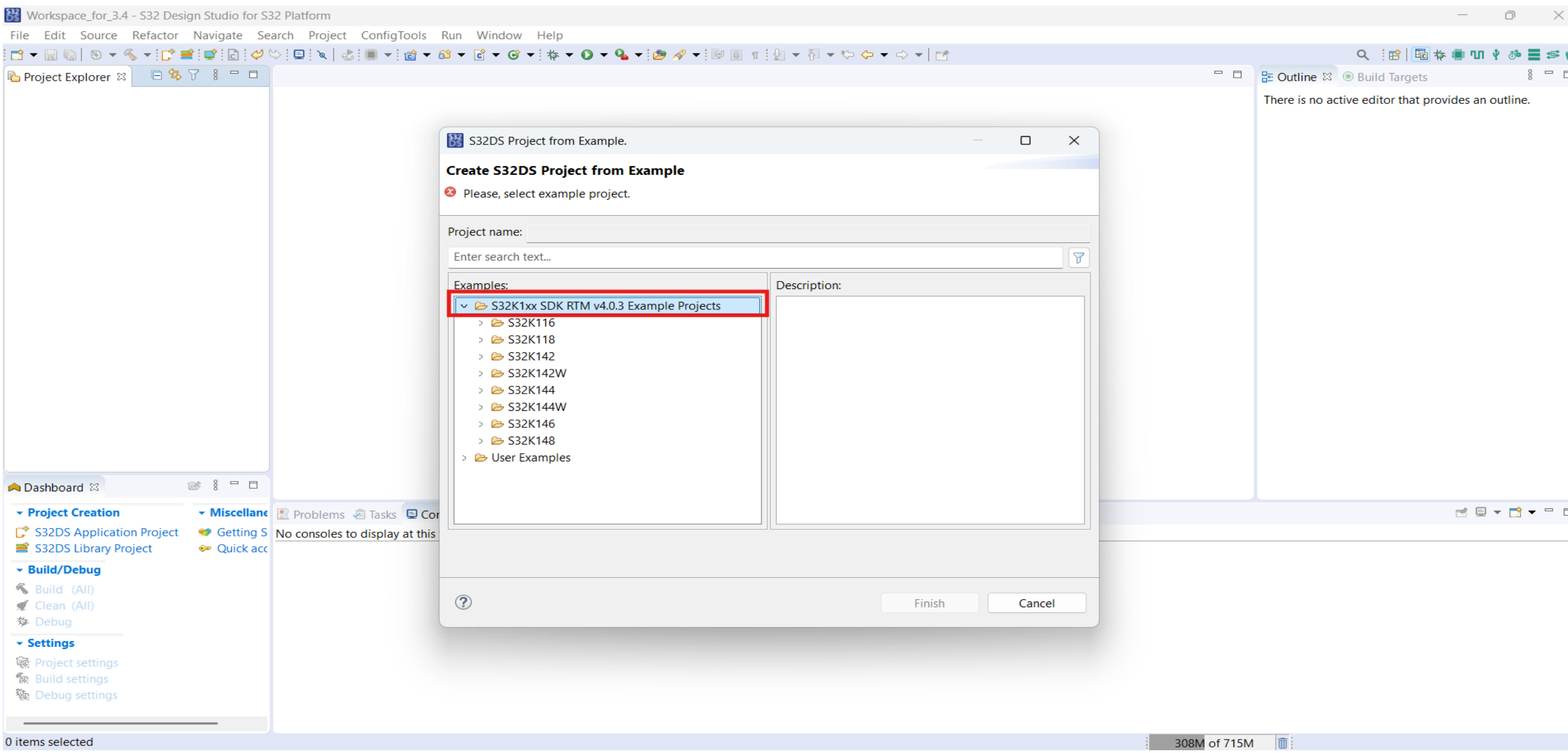
After that a window will pop-up like this, and it will be loading all demo examples from internal SDK that we have installed



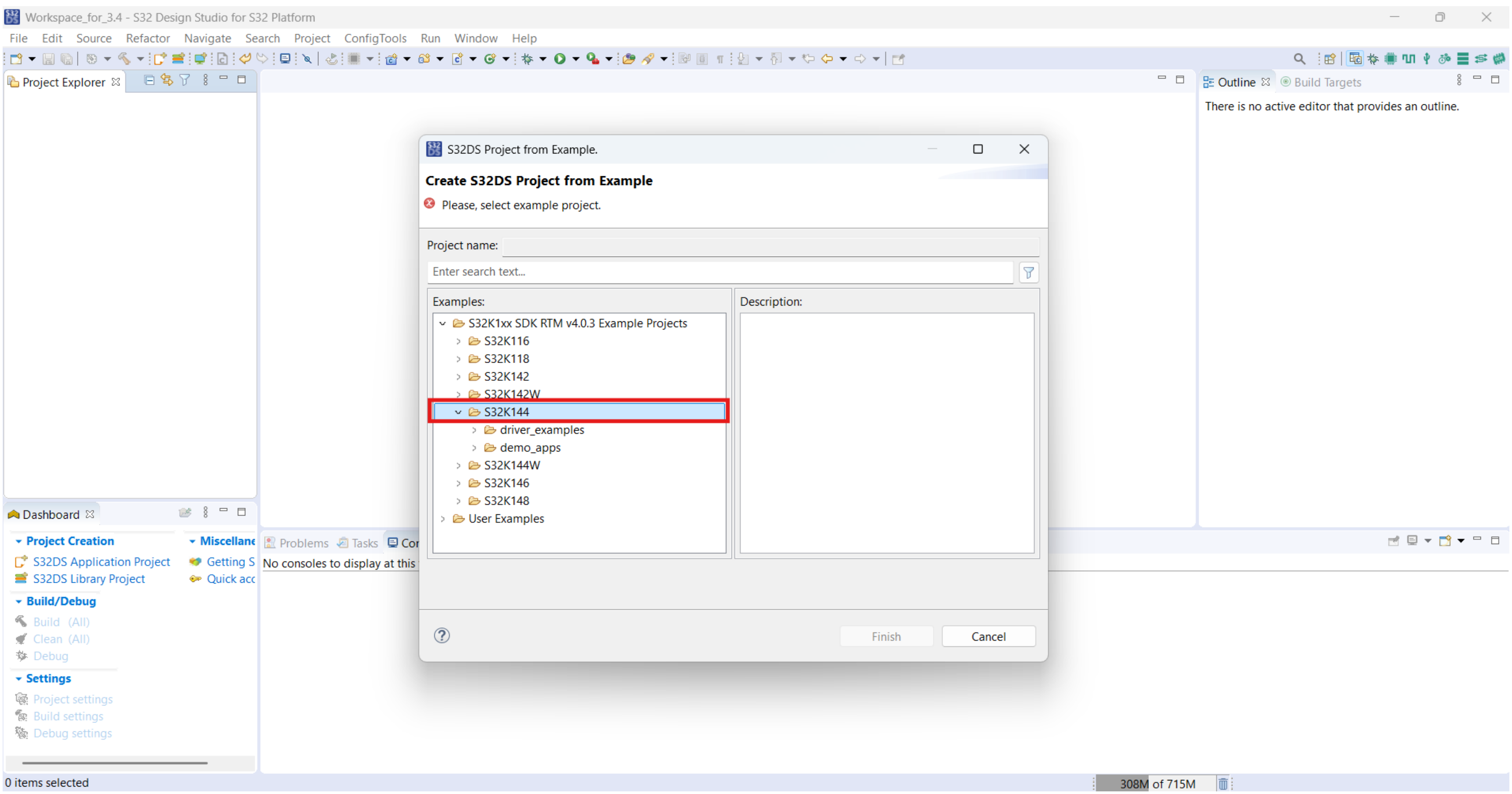
After successful loading, you will see S32K1xx SDK RTM v4.0.3 Example Projects



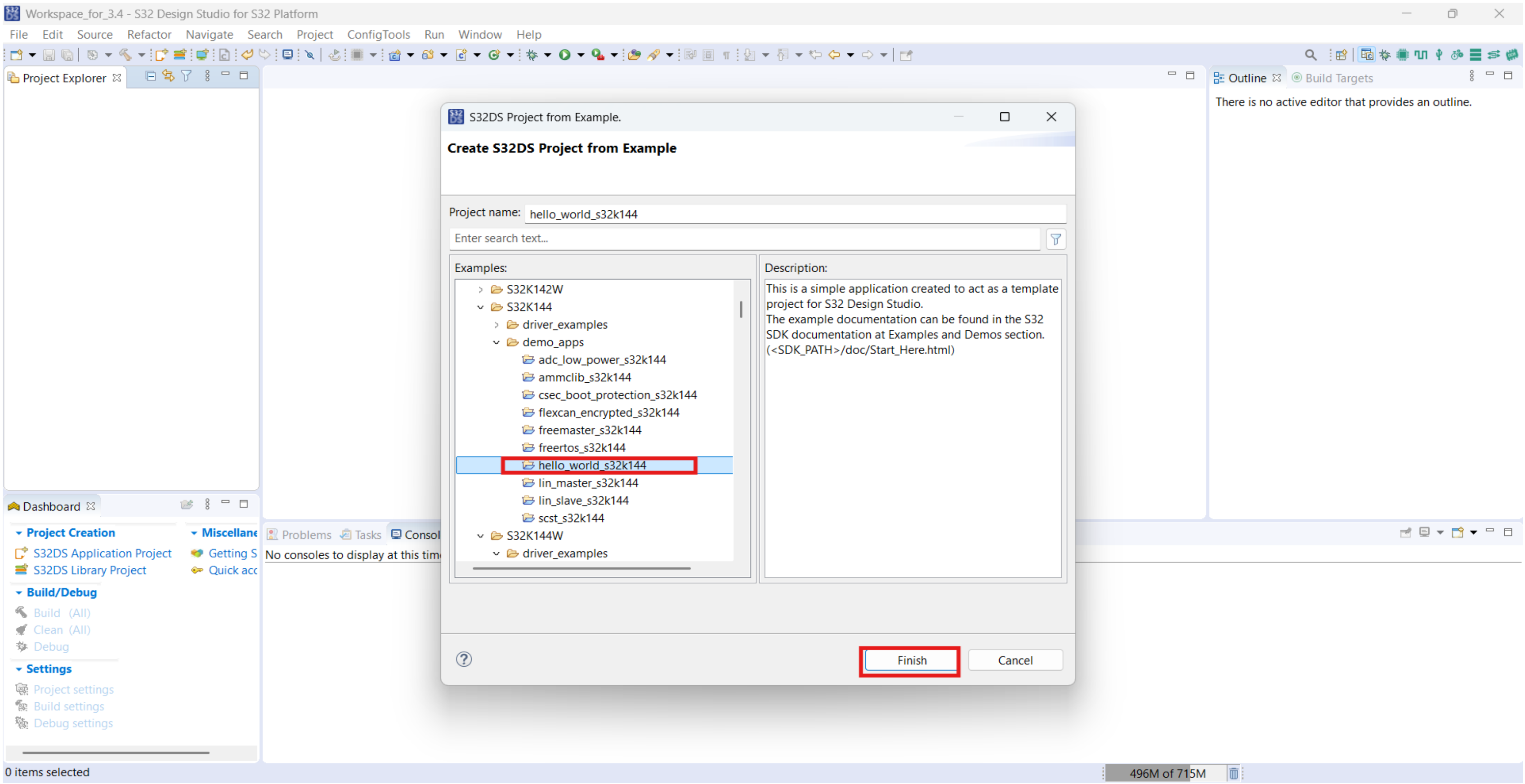
Now click on the SDK name: S32K1xx SDK RTM v4.0.3 Example Projects and we can see list of demo codes for all the MCU's of S32K1 family: S32K116, S32K118, S32K142, S32K142W, S32K144, S32K144W, S32K146, S32K148, as shown below.



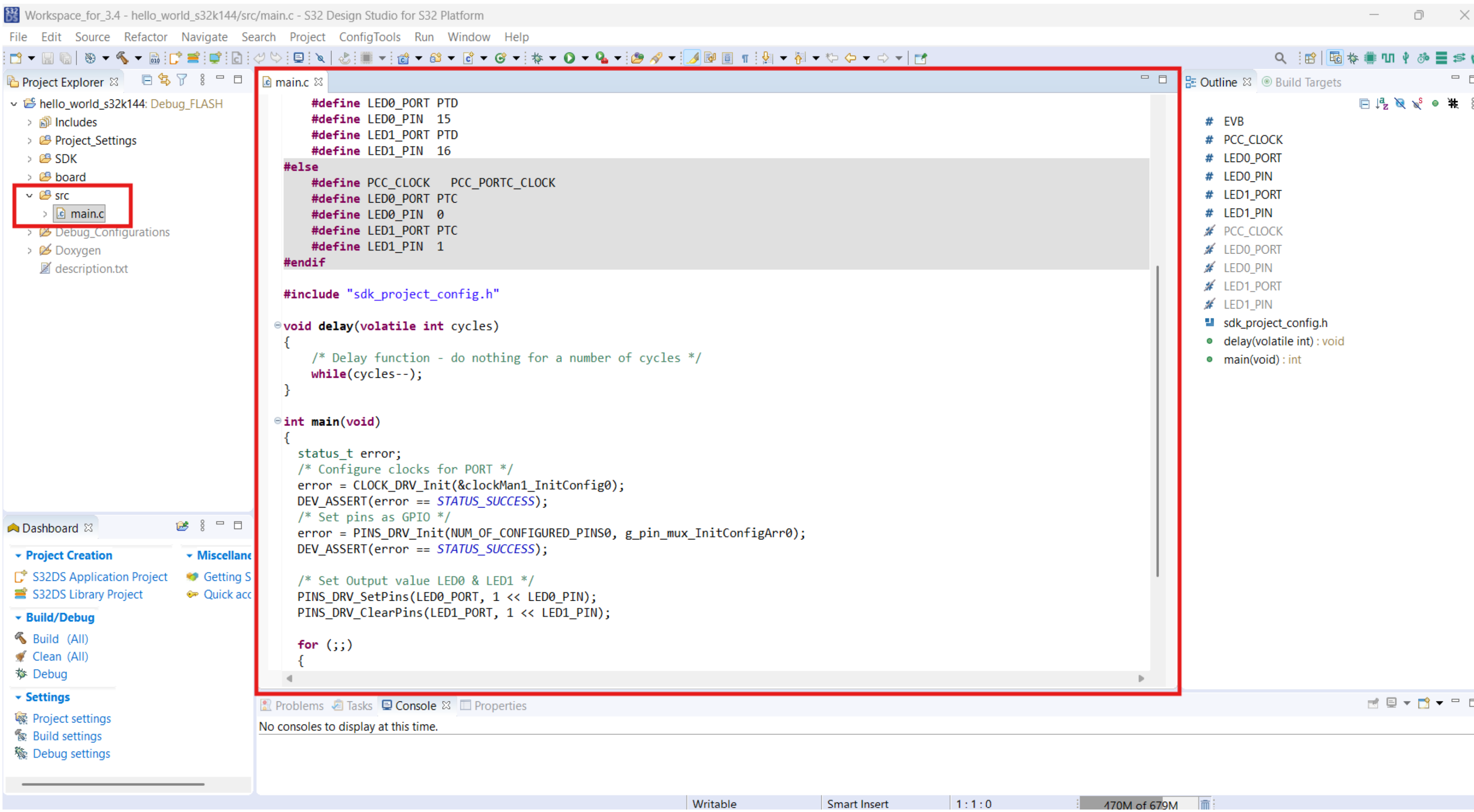
We are using development board which is made on S32K144 MCU, so we will be exploring demo examples of S32K144 MCU.



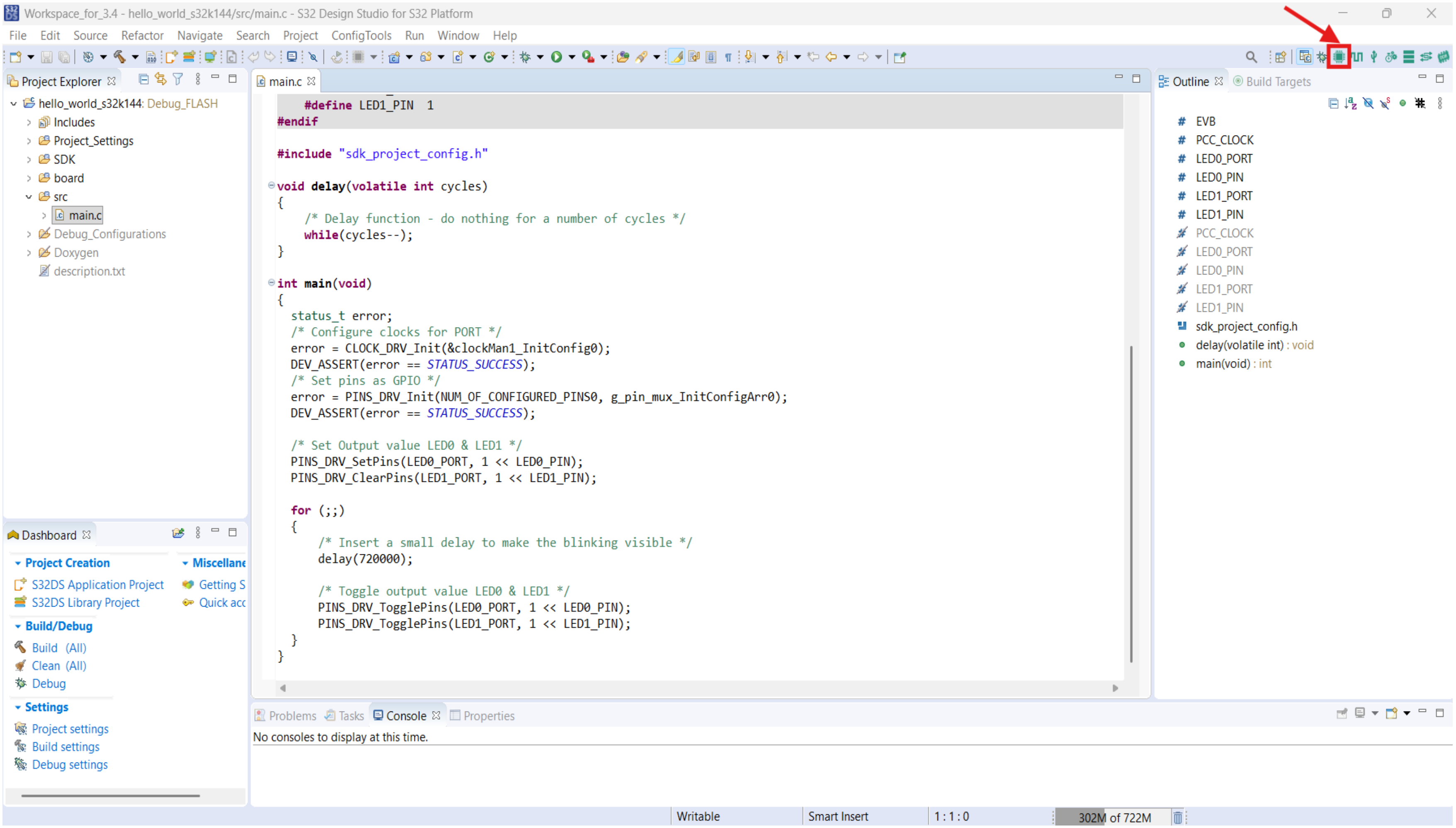
Select hello_world_s32k144 example and click on Finish. This Project blinks the on board RGB LED which is attached at pin PD15 and PD16 of the S32K144 MCU



After opening the project and main.c file, window will look like this:



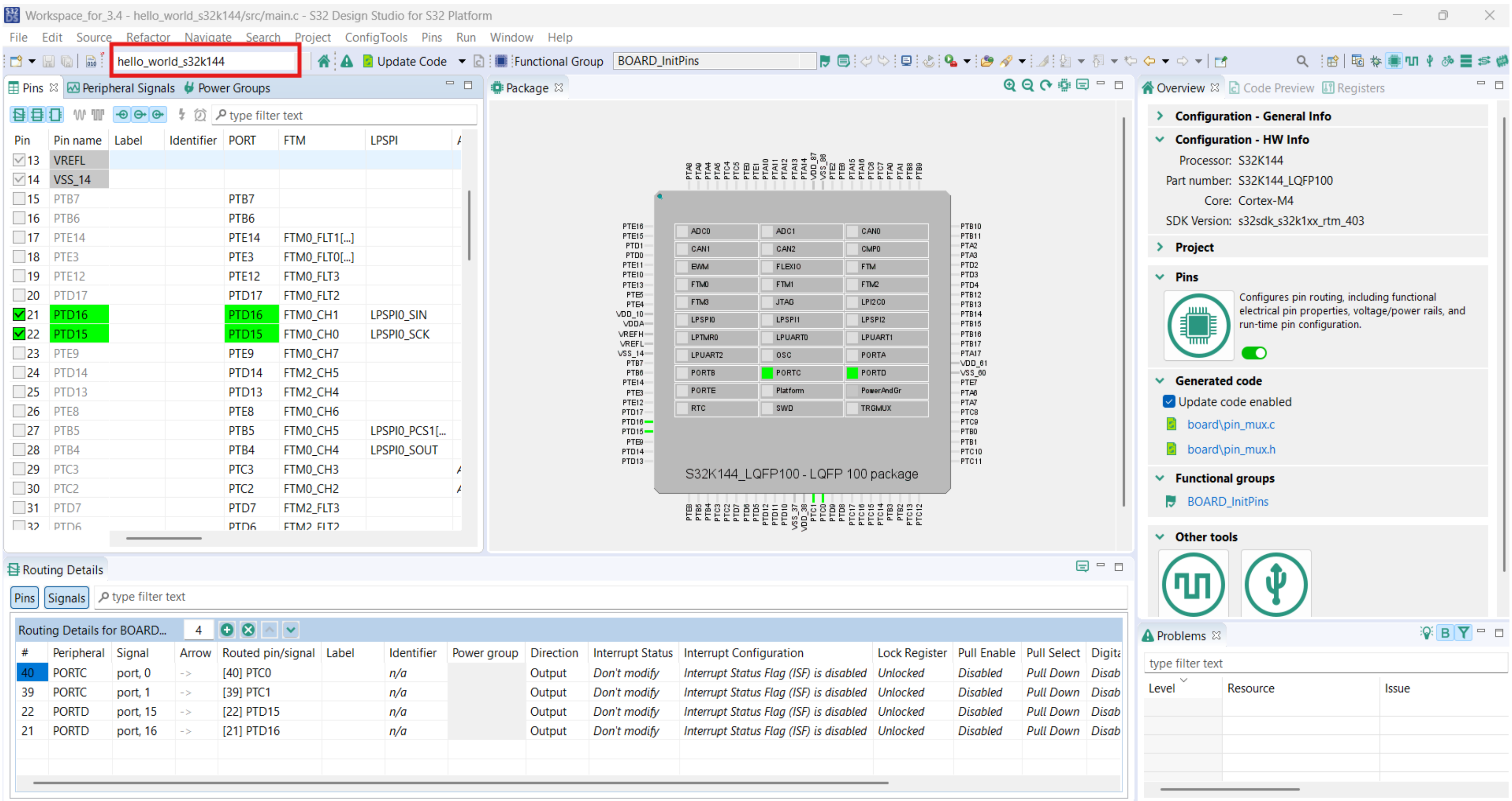
For Pin Routing Details: Click on Pins



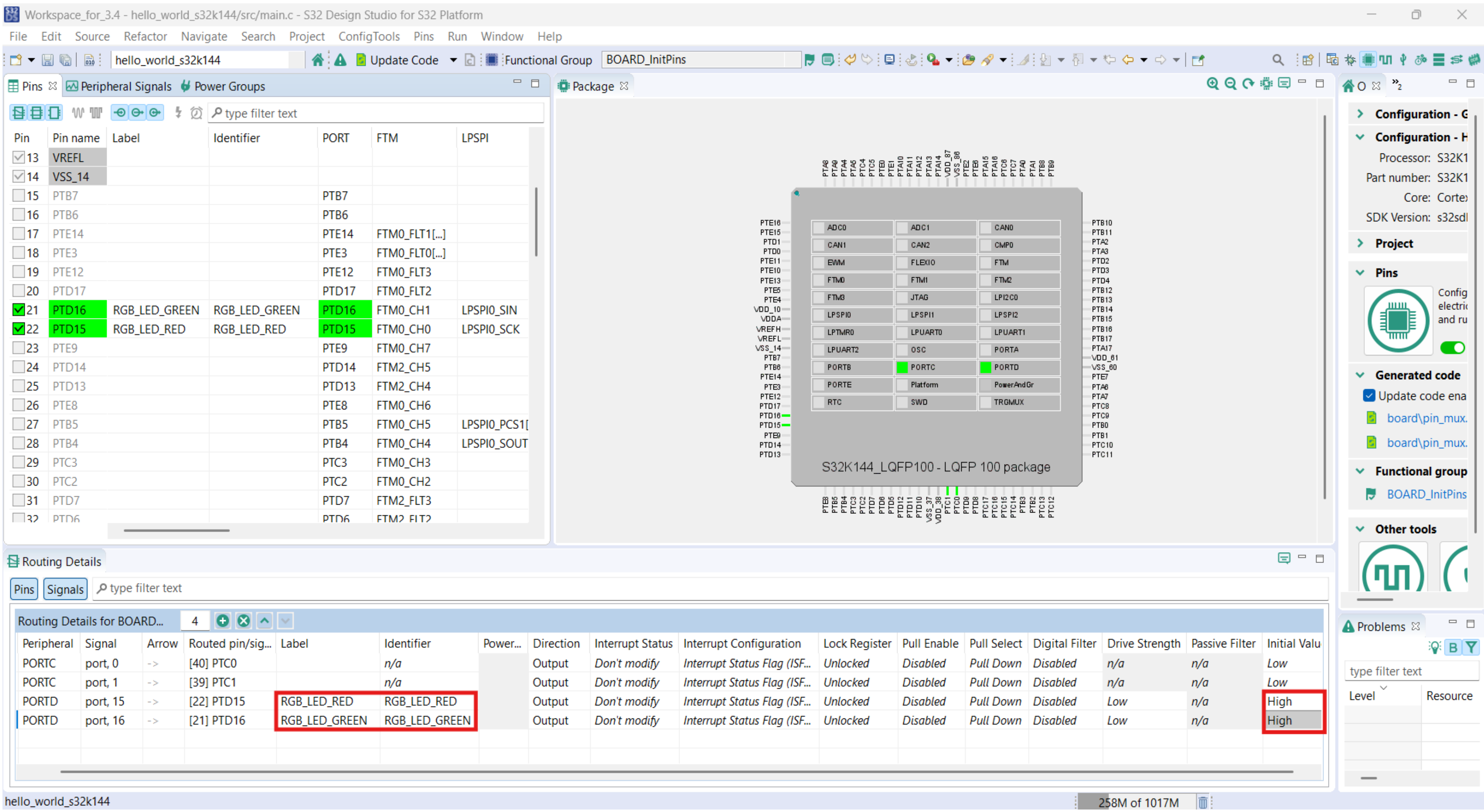
Pinout table

Sl	PIN	Label	Identifier	Description
1	PTD15	RGB_LED_RED	RGB_LED_RED	RGB Red LED, Active Low
2	PTD16	RGB_LED_GREEN	RGB_LED_GREEN	RGB Green LED, Active Low
3	PTD0	RGB_LED_BLUE	RGB_LED_BLUE	RGB Blue LED, Active Low
4	PTC12	USER_SW1	DIG_IP1	User Switch 1, On-board Pulled-up, Active Low
5	PTC13	USER_SW2	DIG_IP2	User Switch 2, On-board Pulled-up, Active Low
6	PTC14	ADC0_CH12	ANG_CH1	Potentiometer 10K ohms, Output Voltage: 0 to 3.3V
7	PTC7	LPUART1_TX	LPUART1_TX	UART Instance-1, Tx
8	PTC6	LPUART1_RX	LPUART1_RX	UART Instance-1, Rx
9	PTA2	LPI2C0_SDA	LPI2C0_SDA	I2C Instance-0, Data pin
10	PTA3	LPI2C0_SCL	LPI2C0_SCL	I2C Instance-0, Clock pin
11	PTE4	CAN0_RX	CAN0_RX	CAN Instance-0, Rx
12	PTE5	CAN0_TX	CAN0_TX	CAN Instance-0, Tx

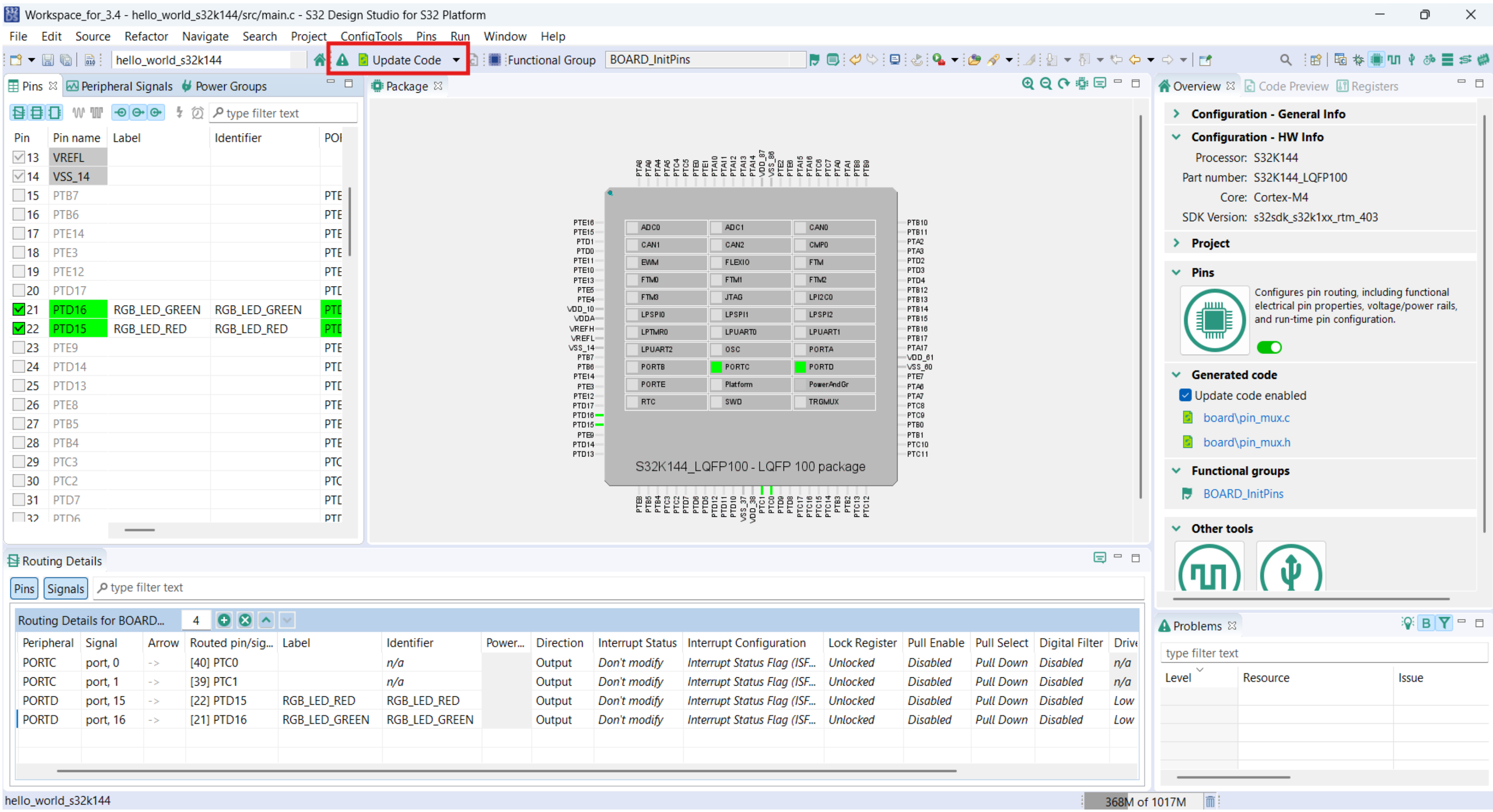
Select hello_world_s32k144 project, it will open ConfigTools for Pins



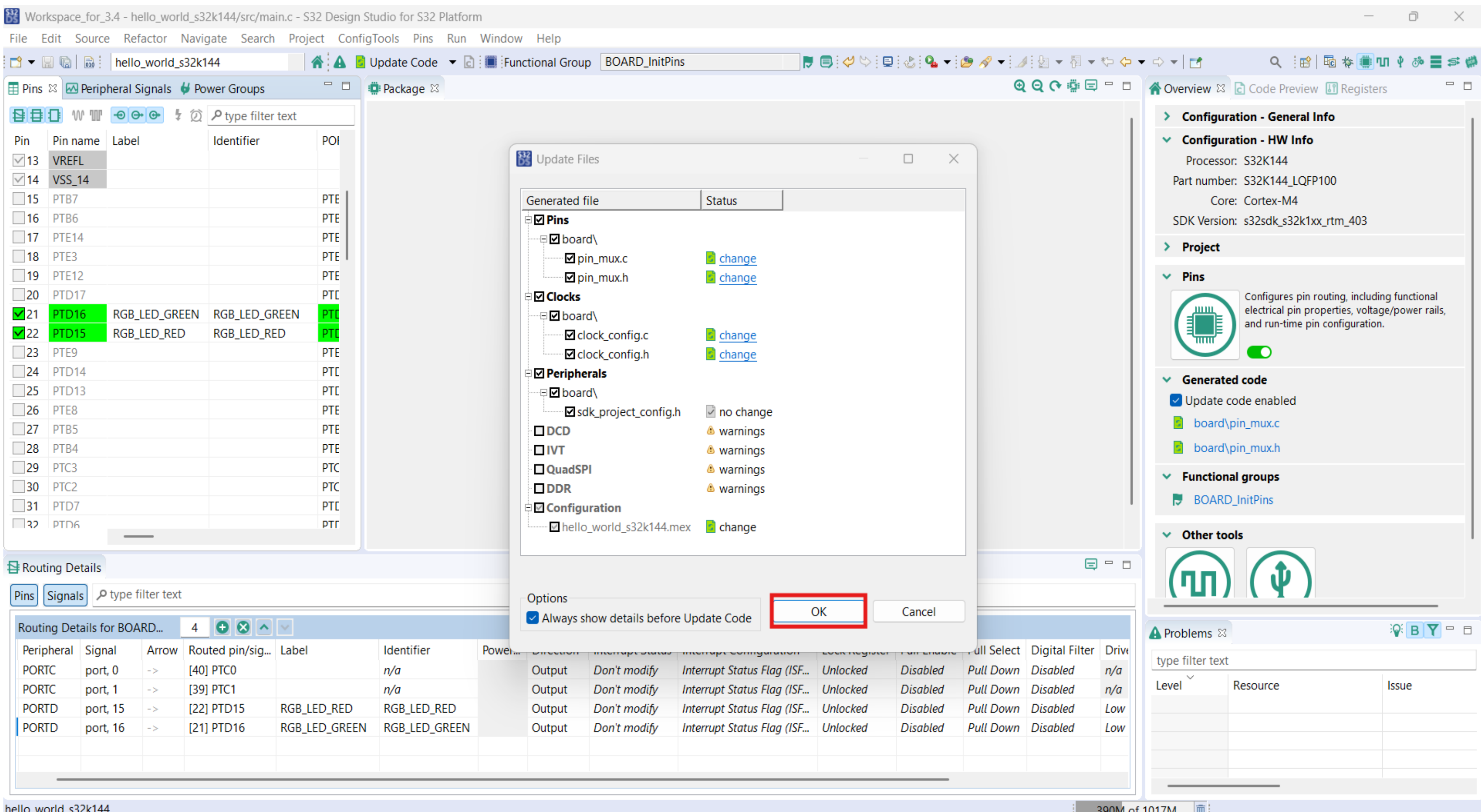
Give name in Label, Identifier and Initial value column as shown below



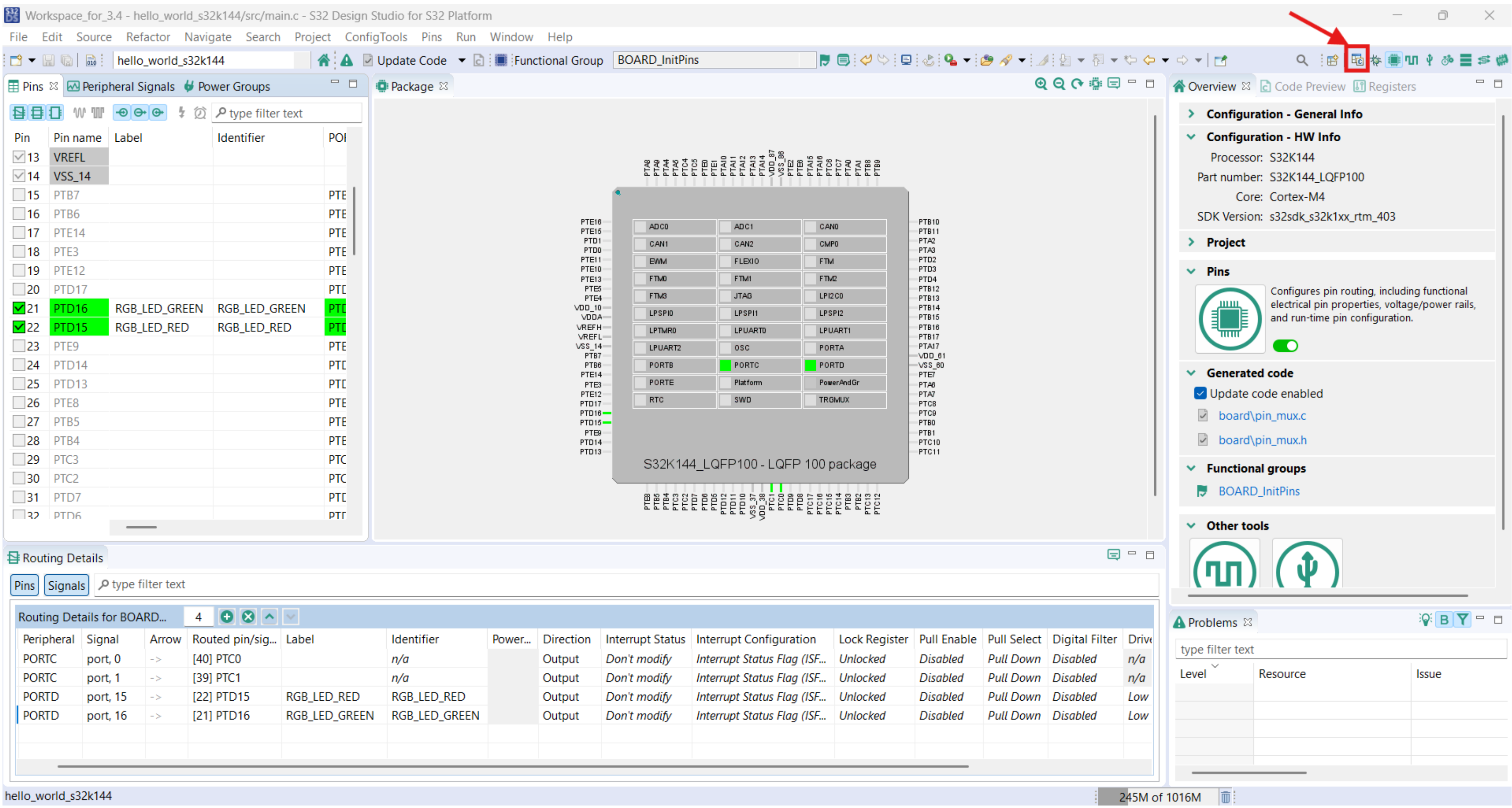
Click on Update Code



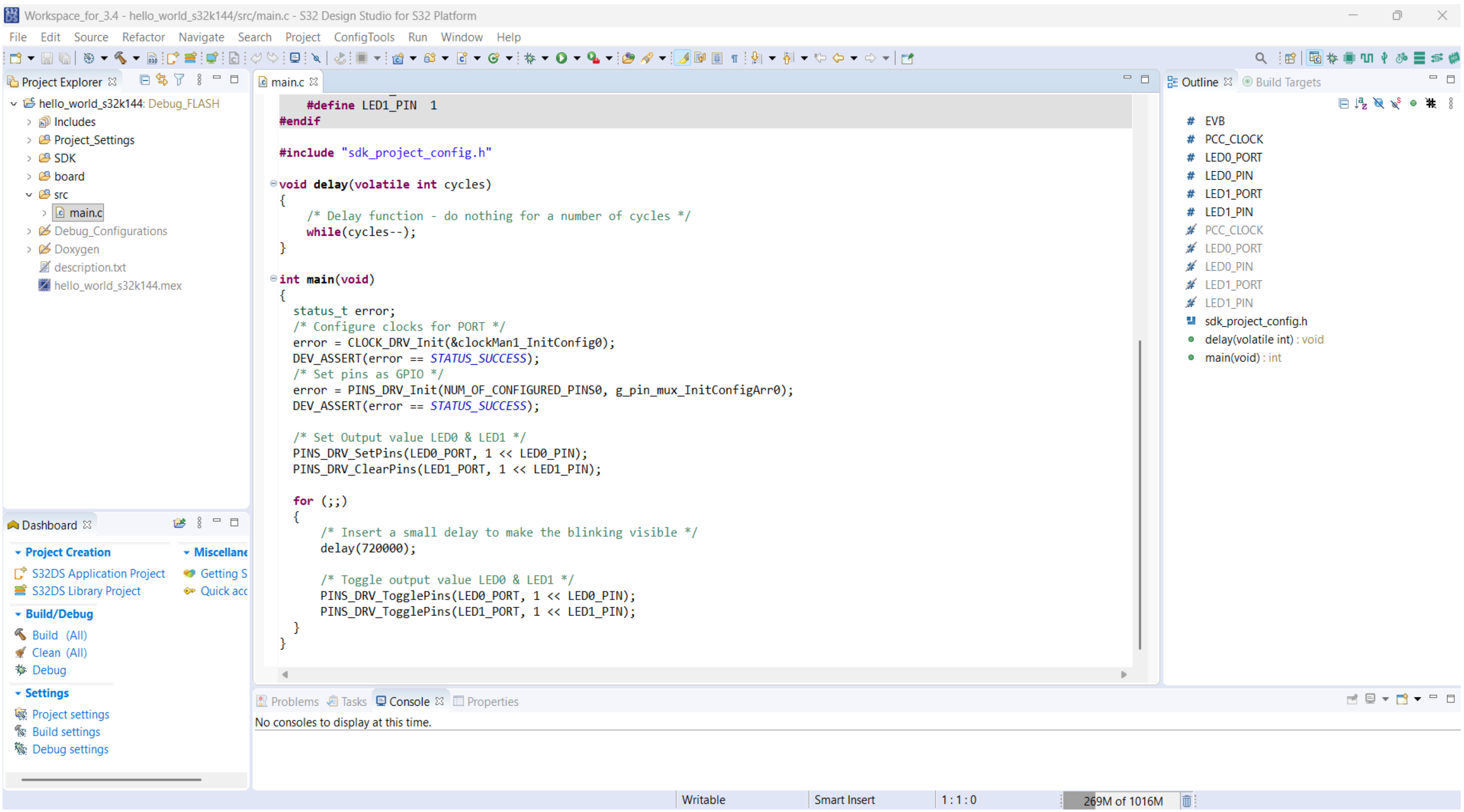
Click on OK


Update Files
Generated file
Status
Pins
board\
pin_mux.c
change
pin_mux.h
change
Clocks
board\
clock_config.c
change
clock_config.h
change
Peripherals
board\
sdk_project_config.h
no change
DCD
warnings
IVT
warnings
QuadSPI
warnings
DDR
warnings
Configuration
hello_world_s32k144.mex
change
Options
Always show details before Update Code
OK
Cancel

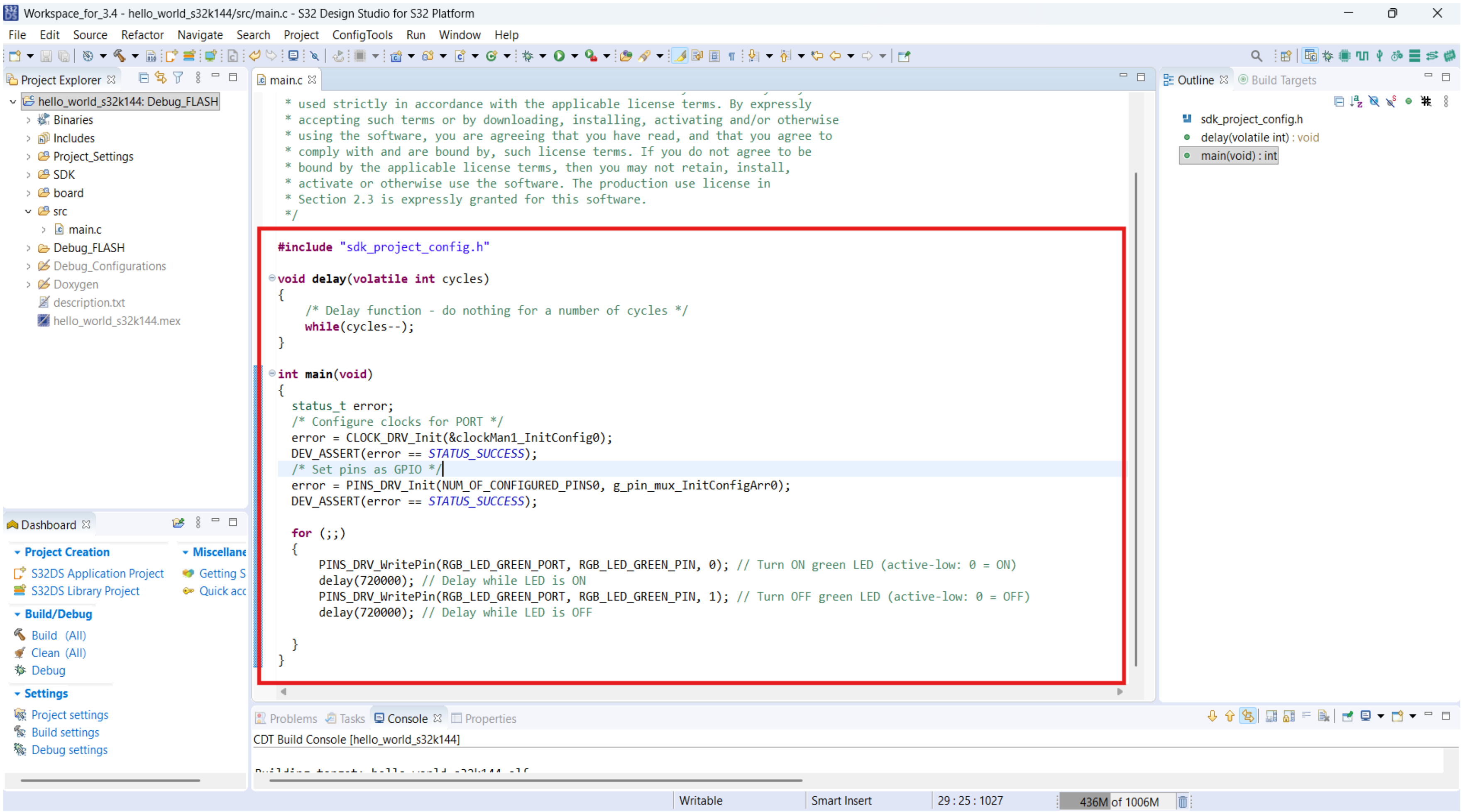
Click on C/C++


Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Pins Run Window Help
hello_world_s32k144
Update Code
Functional Group
BOARD_InitPins
Pins
Peripheral Signals
Power Groups
type filter text
Pin
Pin name
Label
Identifier
13 VREFL
14 VSS_14
15 PTB7
16 PTB6
17 PTE14
18 PTE3
19 PTE12
20 PTD17
21 PTD16 RGB_LED_GREEN RGB_LED_GREEN
22 PTD15 RGB_LED_RED RGB_LED_RED
23 PTE9
24 PTD14
25 PTD13
26 PTE8
27 PTB5
28 PTB4
29 PTC3
30 PTC2
31 PTD7
Package
S32K144_LQFP100 - LQFP 100 package
Overview
Code Preview
Registers
Configuration - General Info
Configuration - HW Info
Processor: S32K144
Part number: S32K144_LQFP100
Core: Cortex-M4
SDK Version: s32sdk_s32k1xx_rtm_403
Project
Pins
Configures pin routing, including functional electrical pin properties, voltage/power rails, and run-time pin configuration.
Generated code
Update code enabled
board\pin_mux.c
board\pin_mux.h
Functional groups
BOARD_InitPins
Other tools
Routing Details
Pins
Signals
Routing Details for BOARD...
Peripheral Signal Arrow Routed pin/sig... Label Identifier Power... Direction Interrupt Status Interrupt Configuration Lock Register Pull Enable Pull Select Digital Filter
PORTC port, 0 -> [40] PTC0 n/a Output Don't modify Interrupt Status Flag (ISF... Unlocked Disabled Pull Down Disabled
PORTC port, 1 -> [39] PTC1 n/a Output Don't modify Interrupt Status Flag (ISF... Unlocked Disabled Pull Down Disabled
PORTD port, 15 -> [22] PTD15 RGB_LED_RED RGB_LED_RED Output Don't modify Interrupt Status Flag (ISF... Unlocked Disabled Pull Down Disabled
PORTD port, 16 -> [21] PTD16 RGB_LED_GREEN RGB_LED_GREEN Output Don't modify Interrupt Status Flag (ISF... Unlocked Disabled Pull Down Disabled
Problems
Level
Resource
Issue
hello_world_s32k144
245M of 1016M

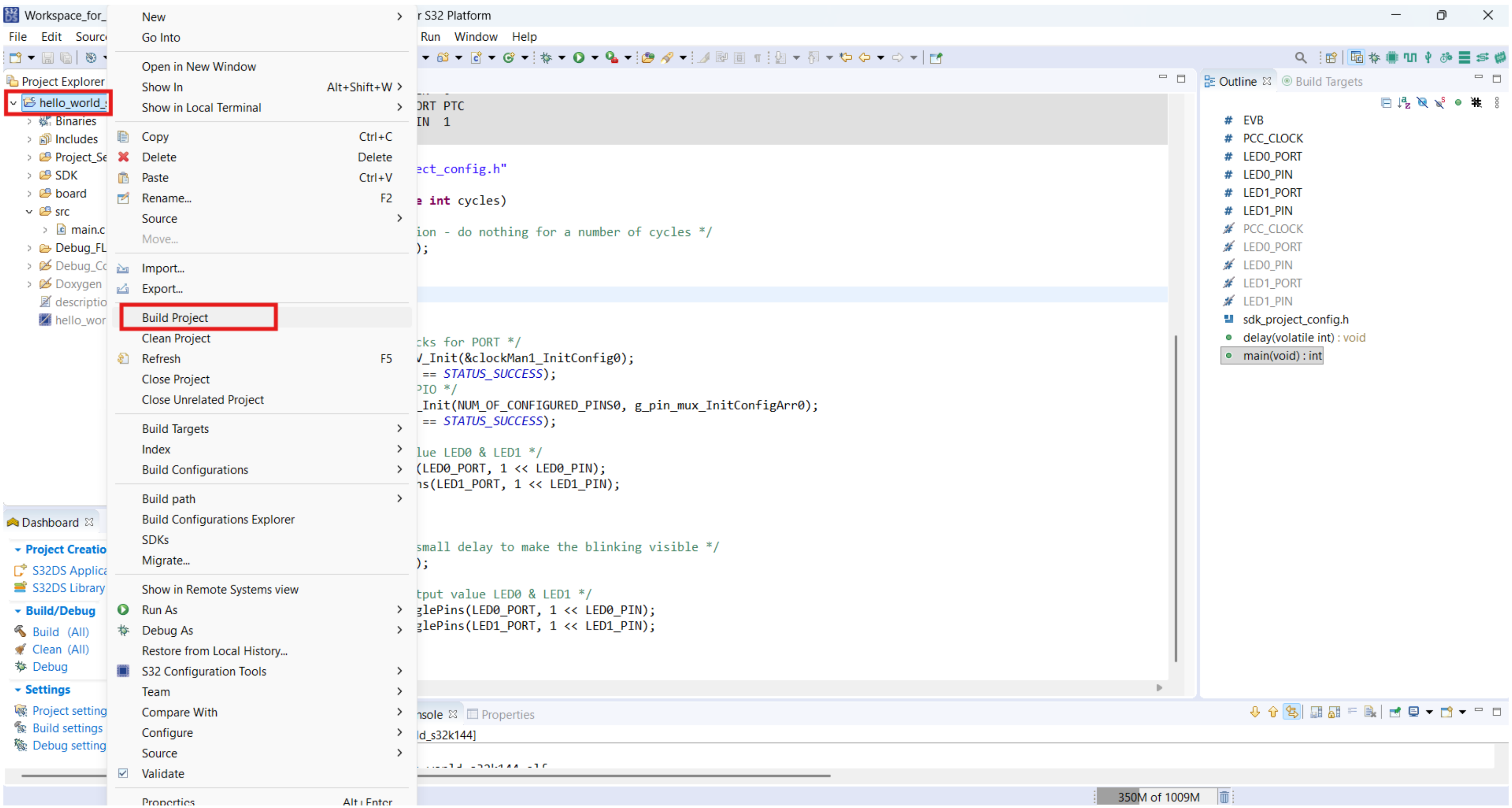
It will open C/C++ perspective



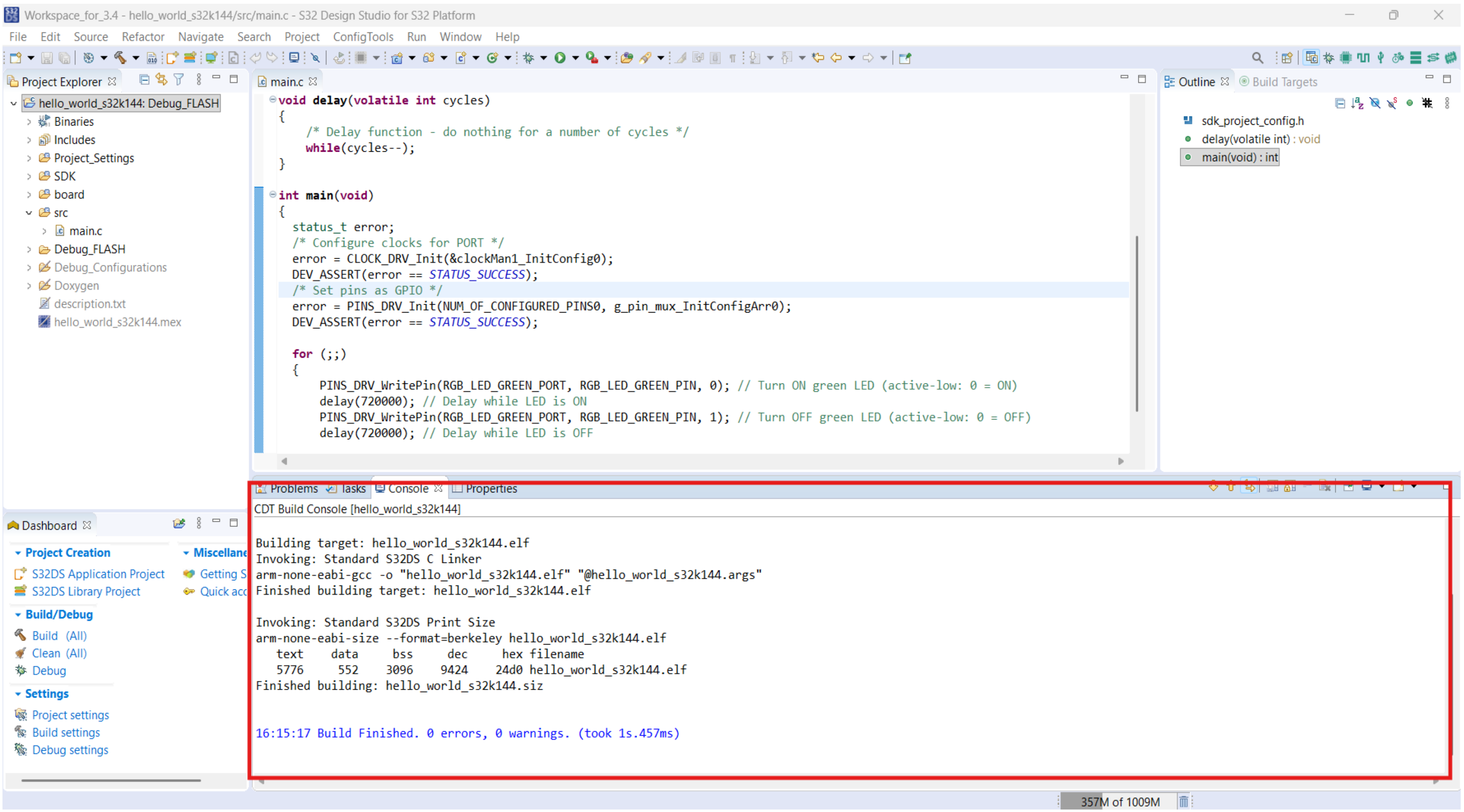
Write code and save project



Select project and right click and click on Build project

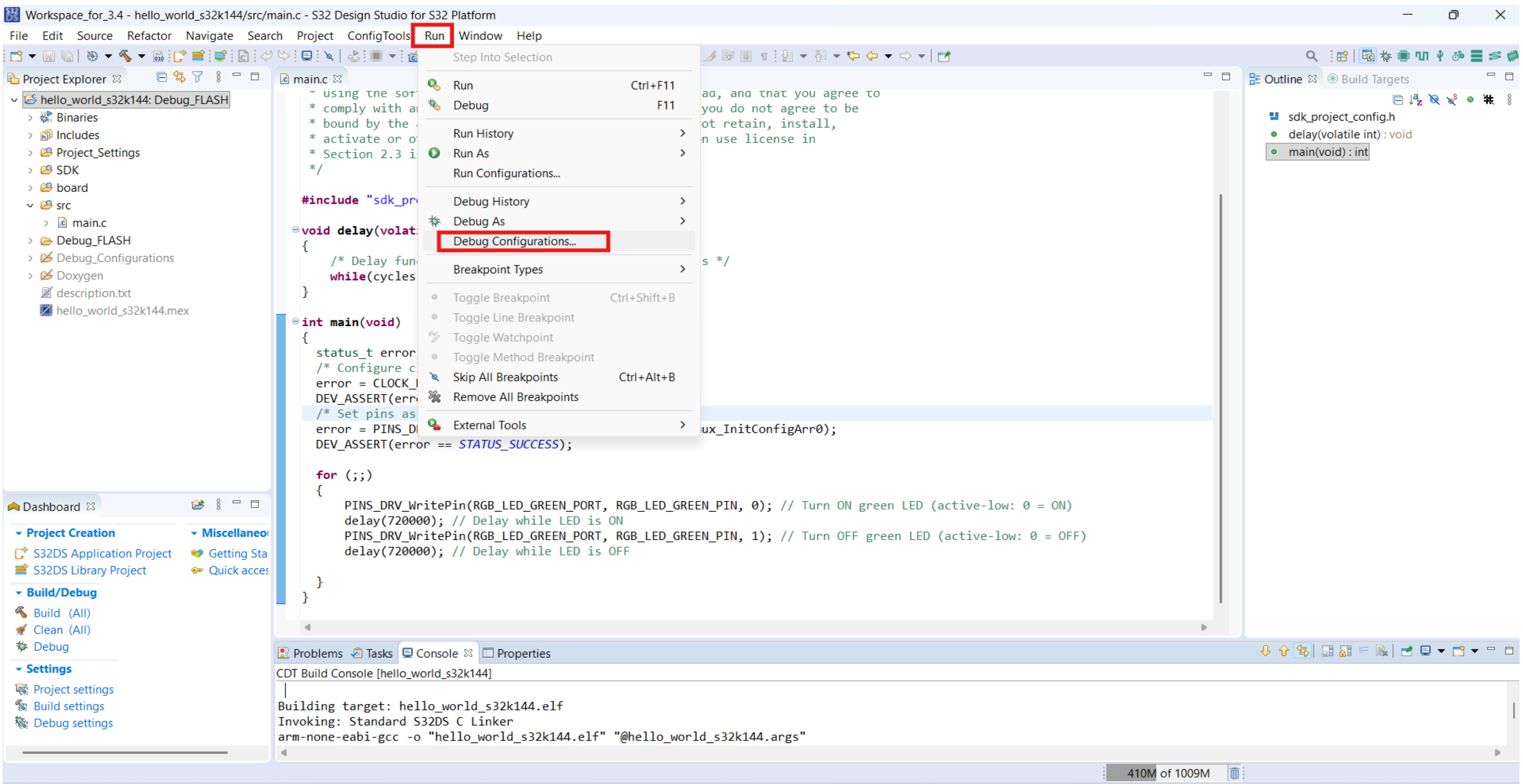


Check for any errors

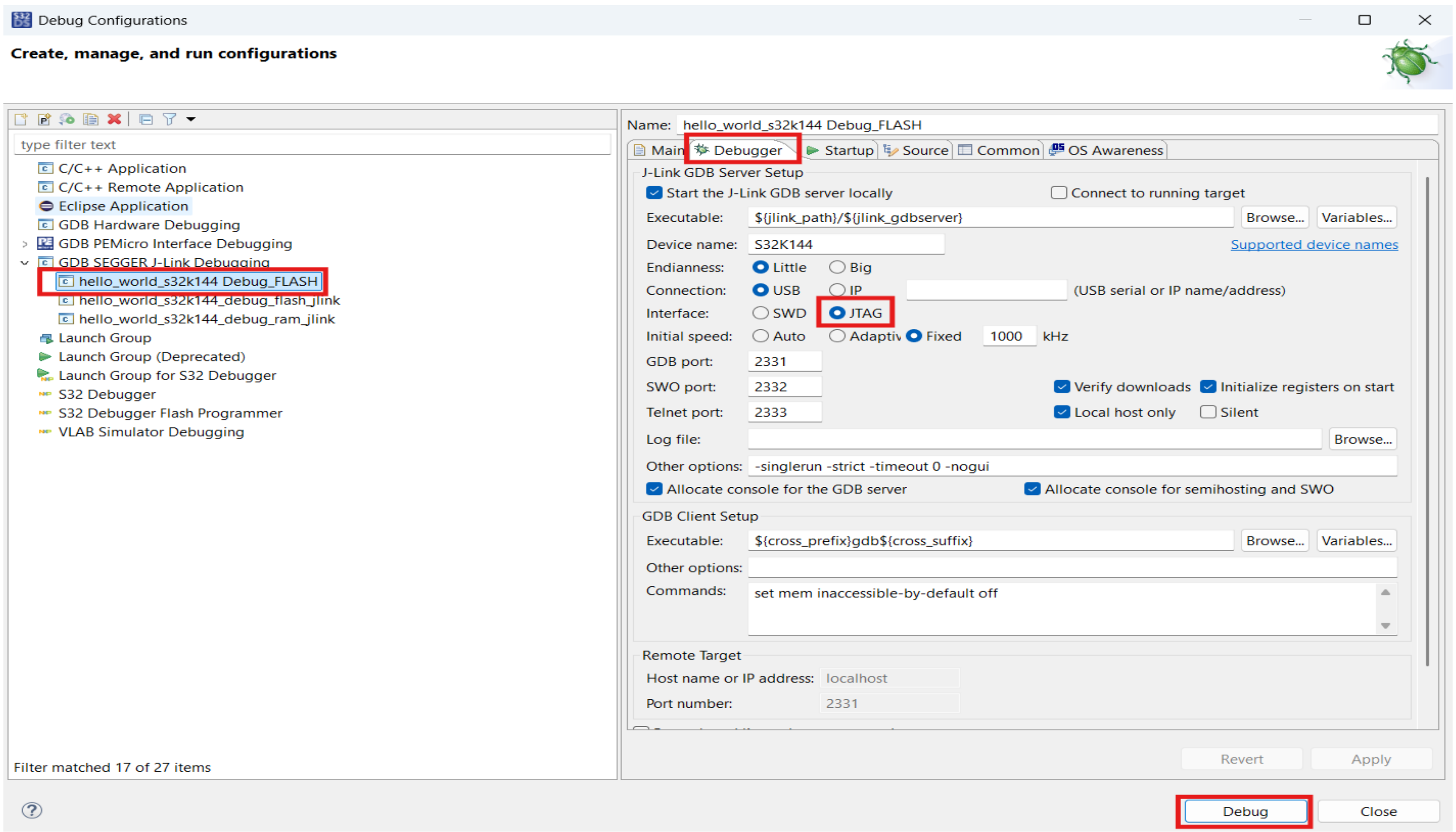


Connect S32K144 based development board to laptop/PC.

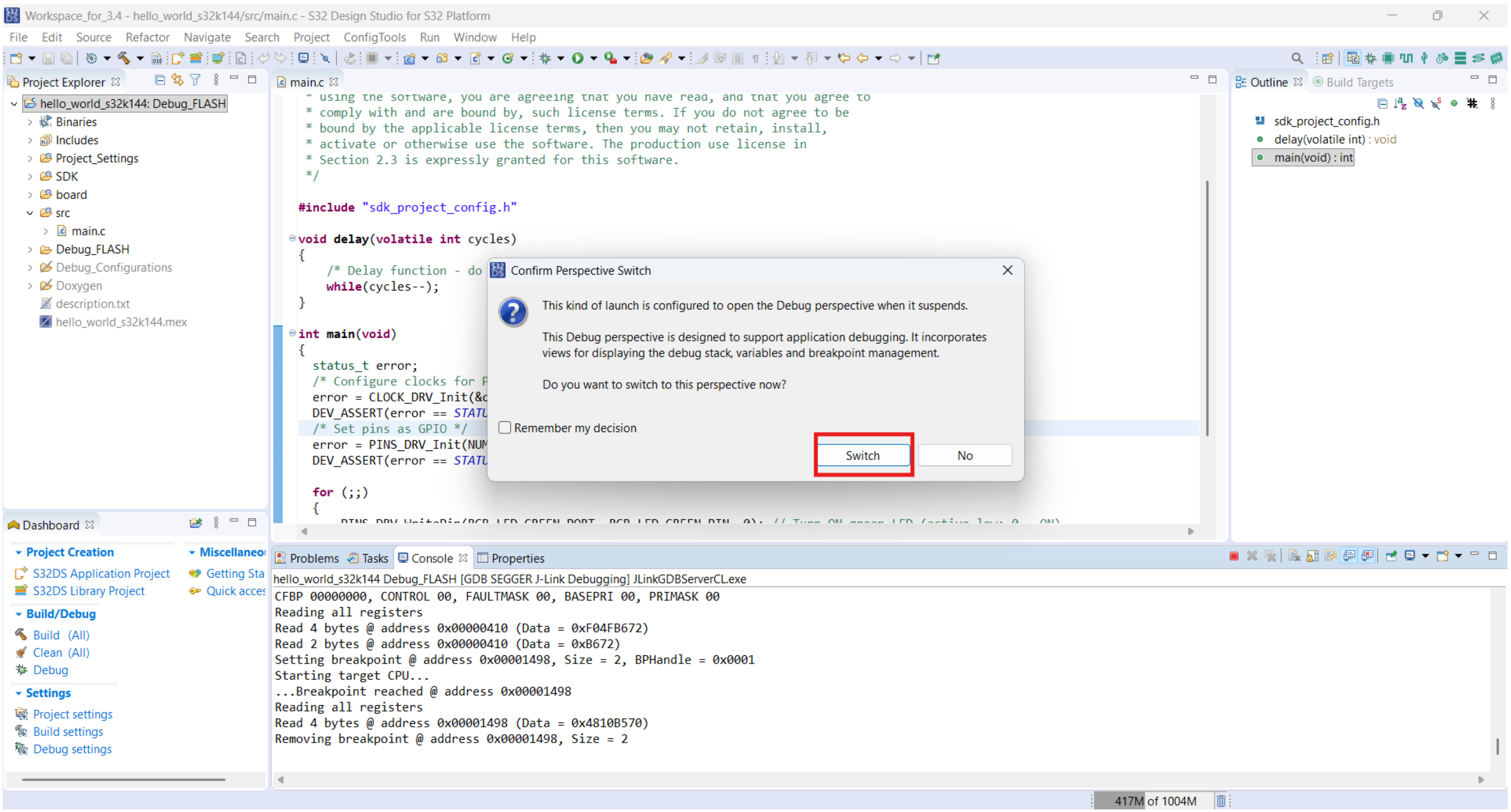
Flash using debugger: Run -> Debug Configuration



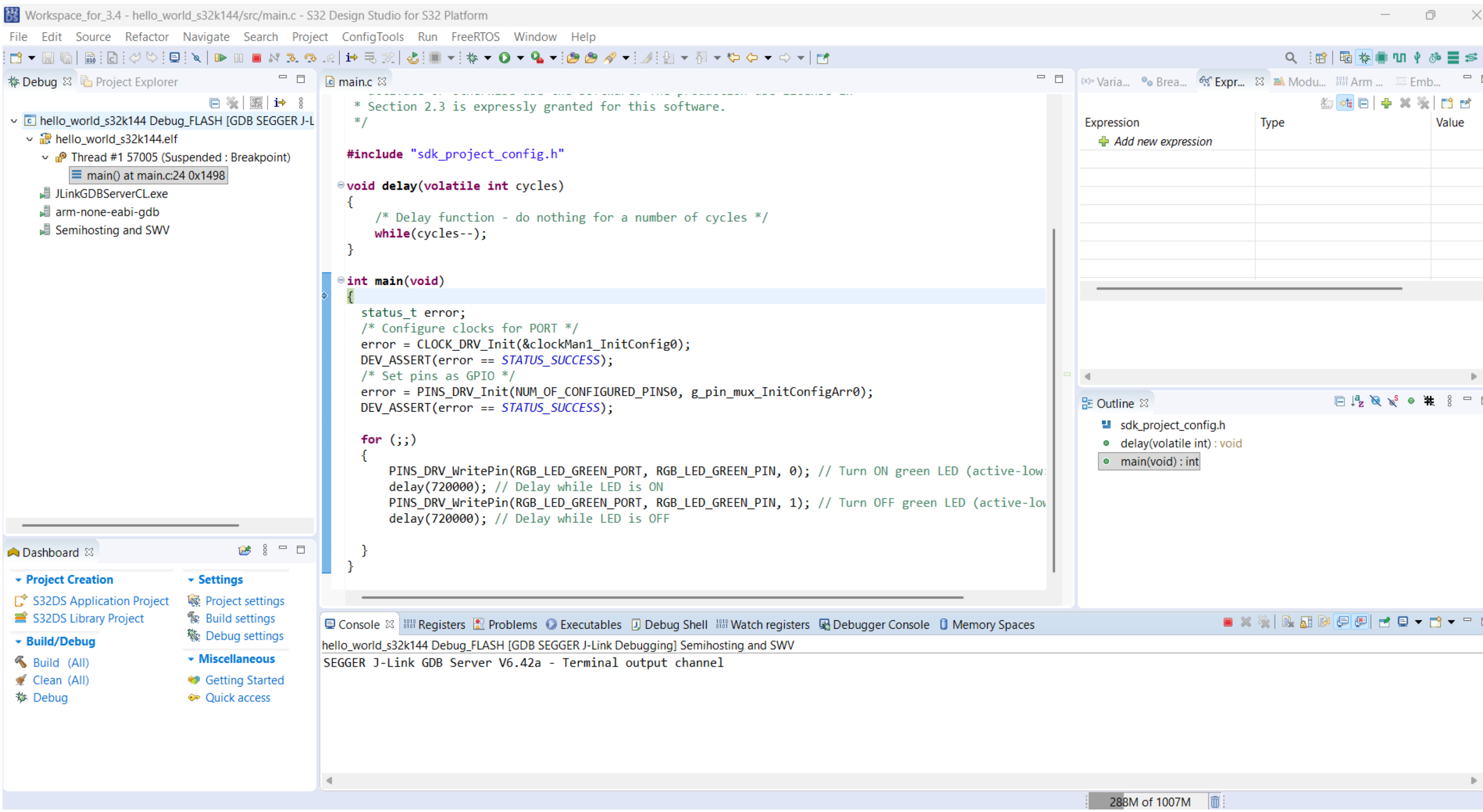
Click on Debugger and do settings as below and Click on Debug



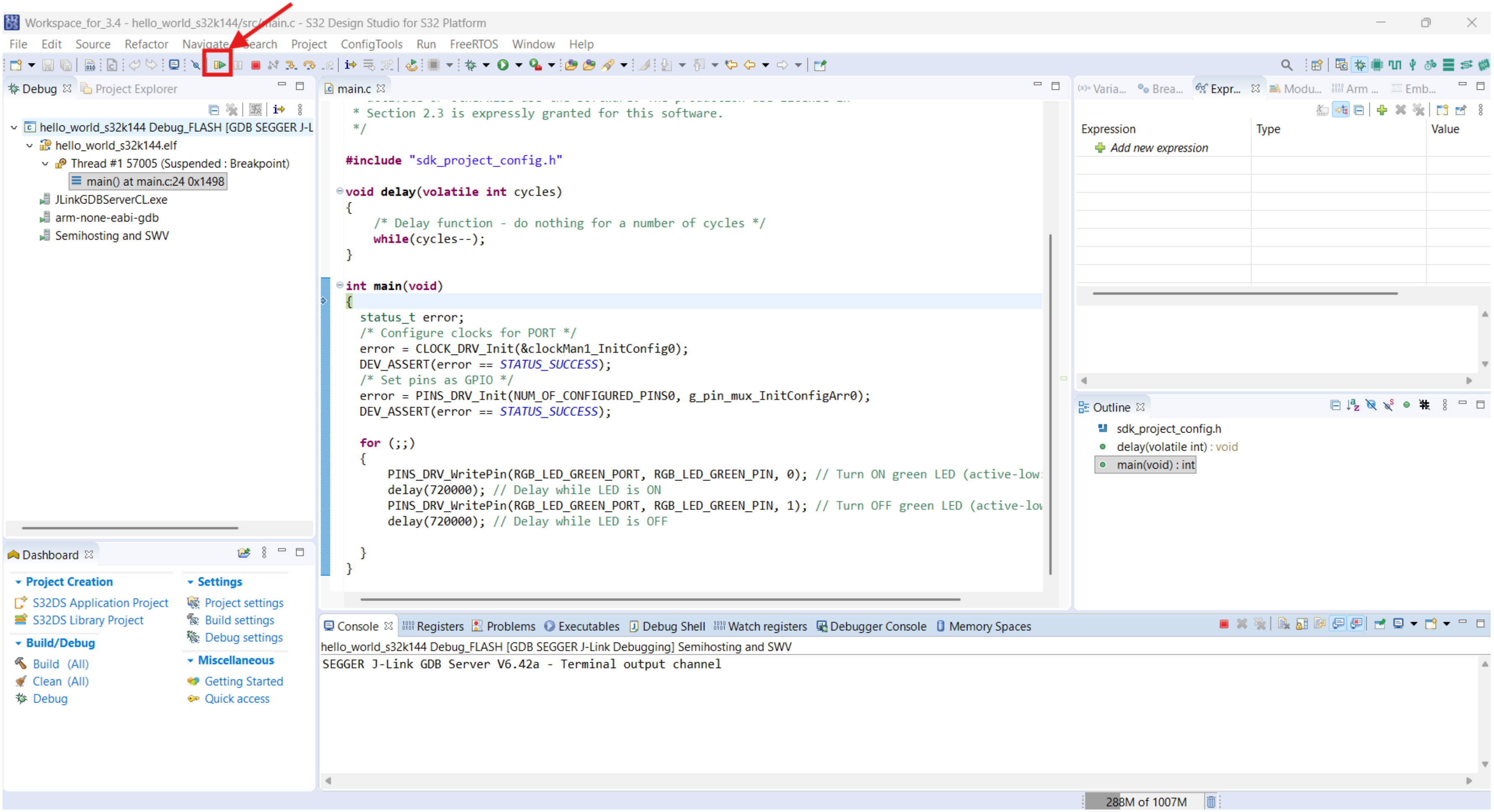
Click on Switch



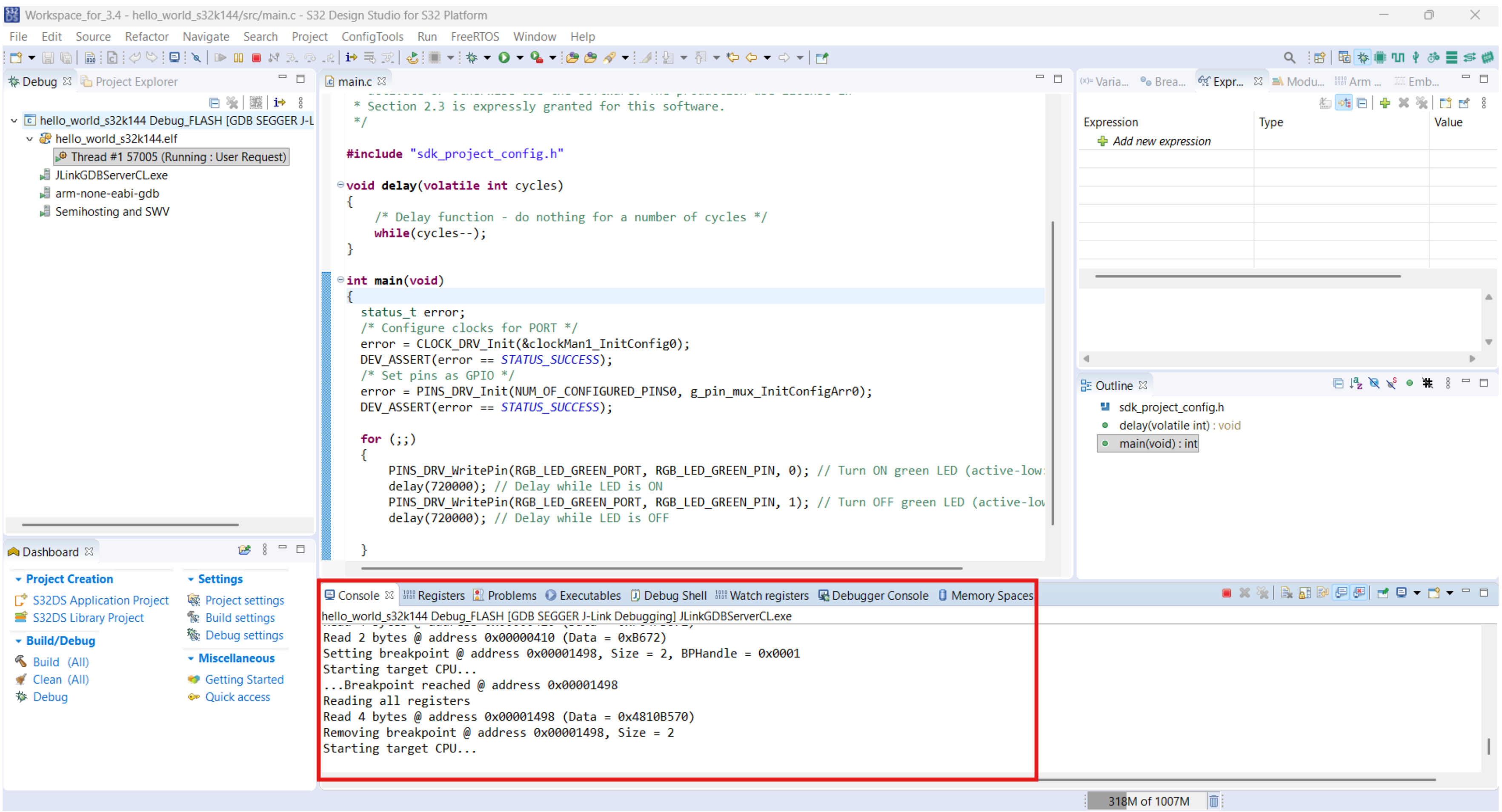
Open Debug perspective



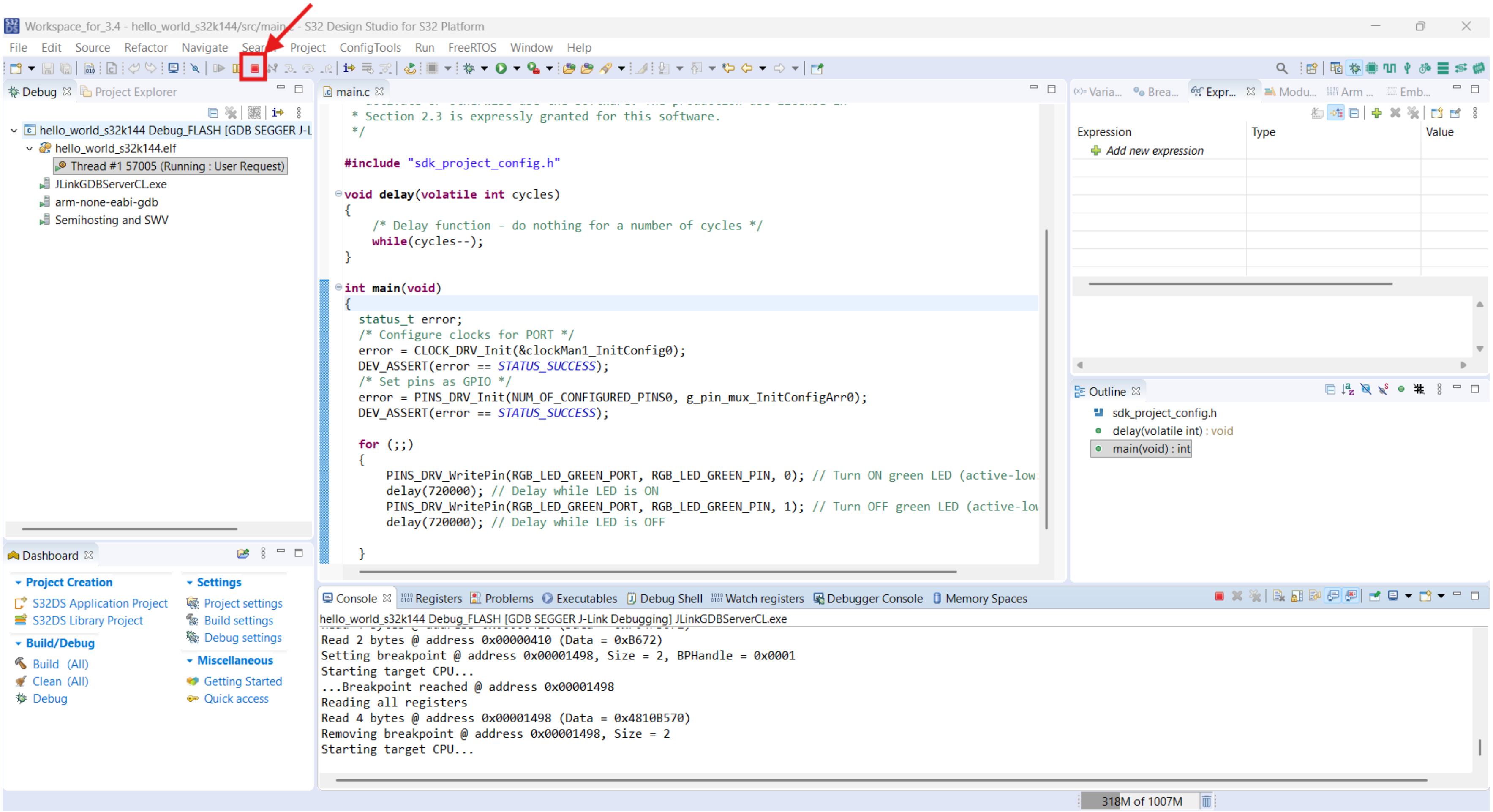
Click on Resume, starts debugging



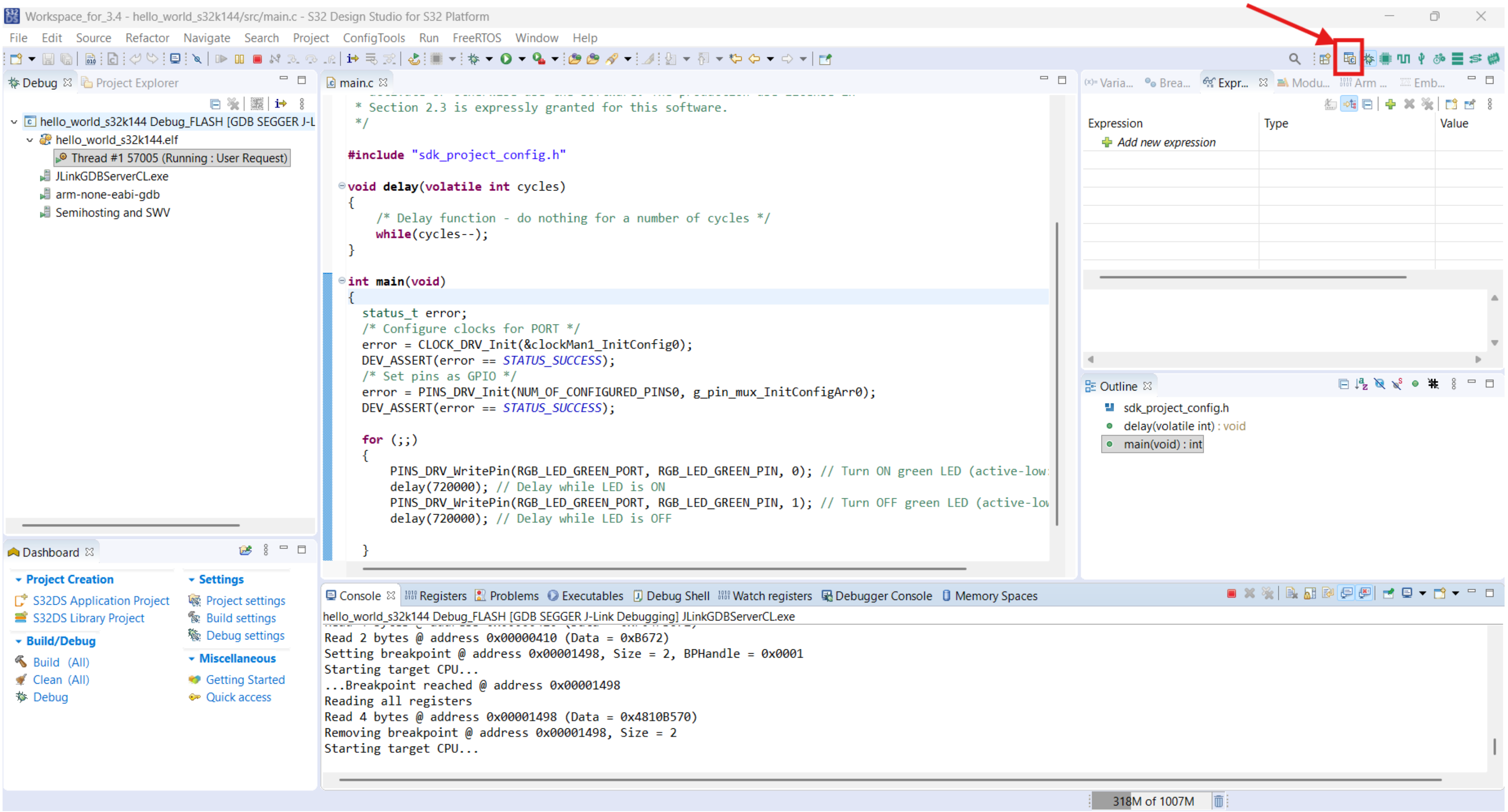
Check the output on connected development board



Click on Terminate to stop debugging

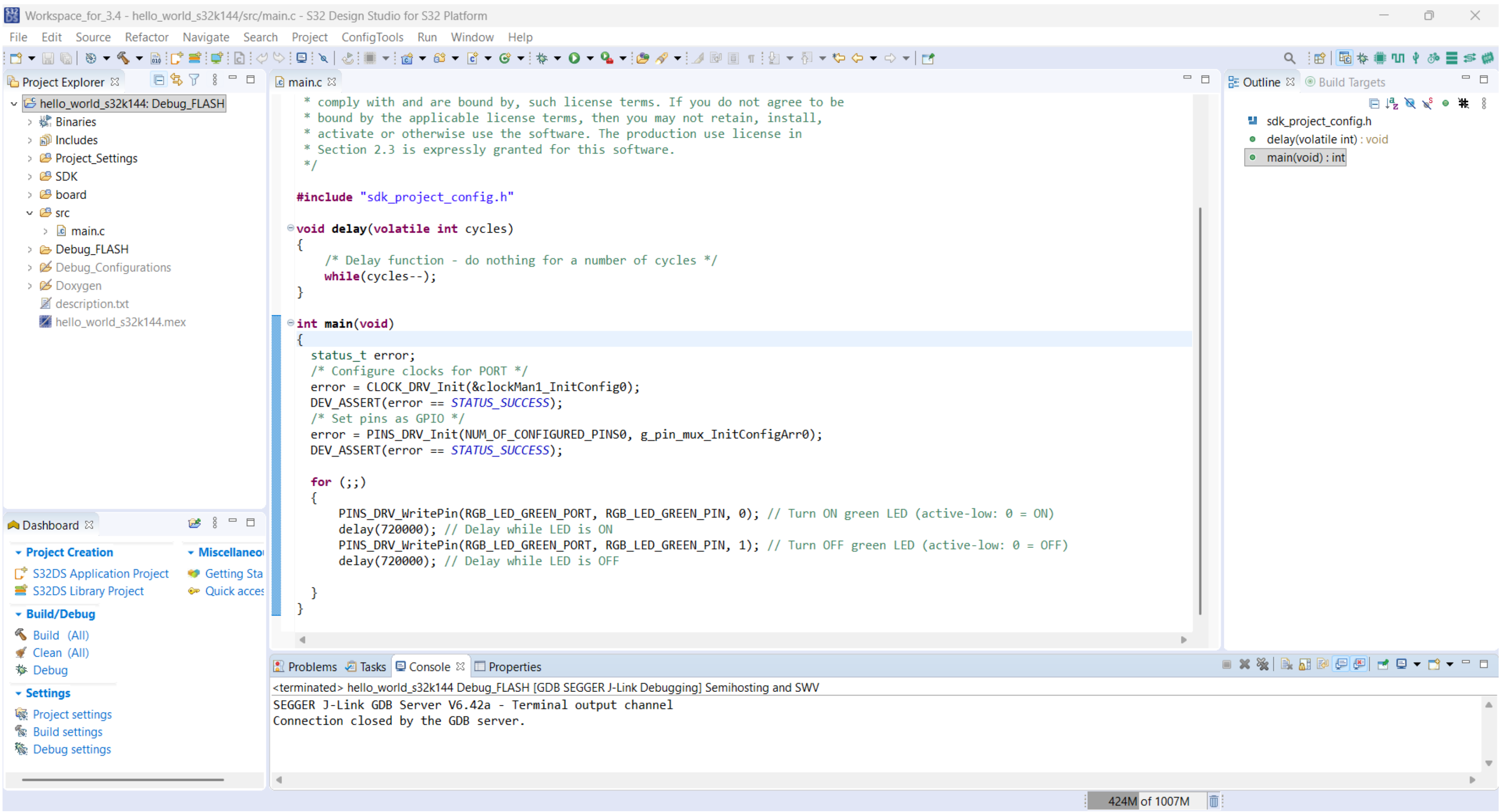


Click on C/C++

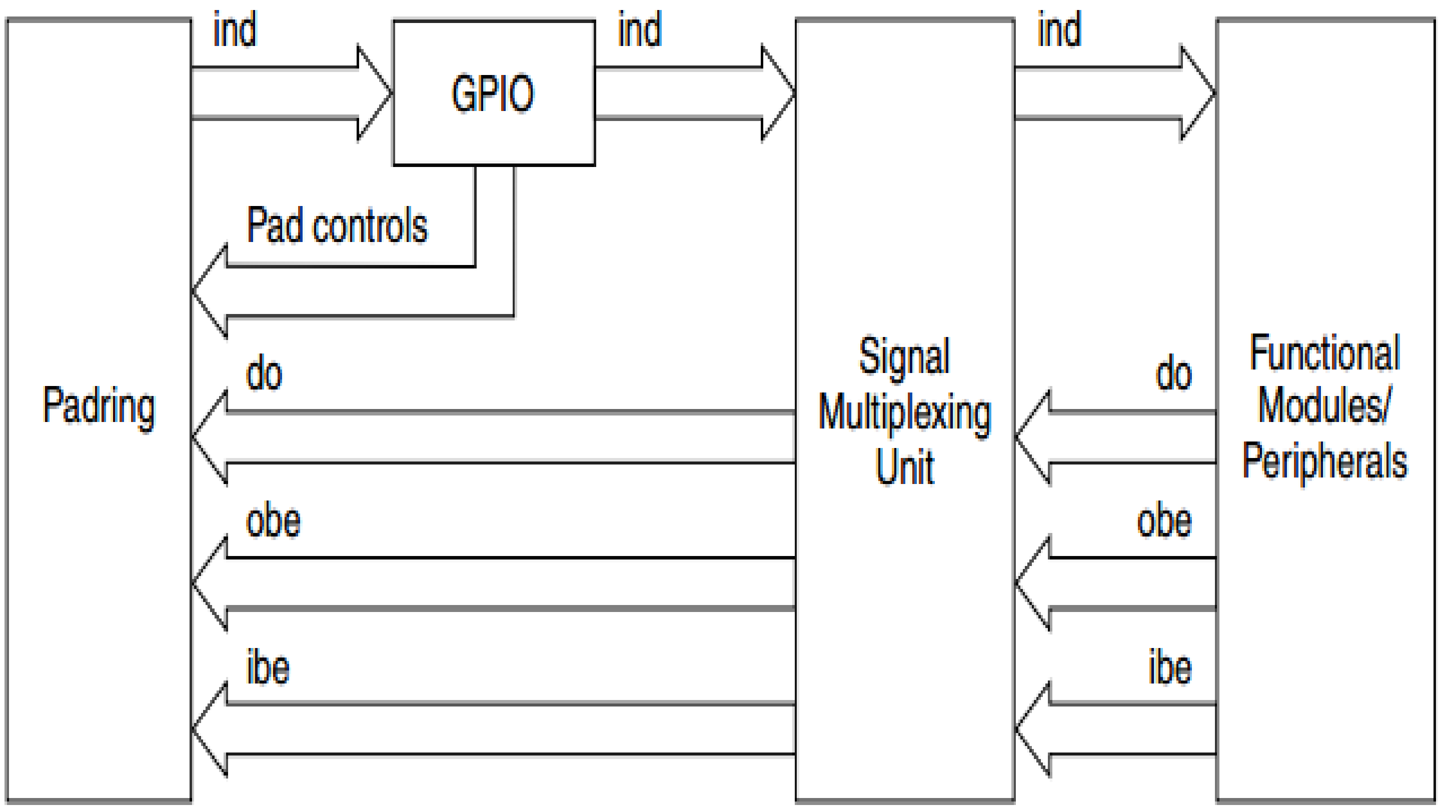


Workspace_for_3.4 - hello_world_s32k144/src/main.c - S32 Design Studio for S32 Platform
File Edit Source Refactor Navigate Search Project ConfigTools Run FreeRTOS Window Help
Debug
Project Explorer
hello_world_s32k144 Debug_FLASH [GDB SEGGER J-L
hello_world_s32k144.elf
Thread #1 57005 (Running : User Request)
JLinkGDBServerCL.exe
arm-none-eabi-gdb
Semihosting and SWV
main.c
* Section 2.3 is expressly granted for this software.
*/
#include "sdk_project_config.h"
void delay(volatile int cycles)
{
/* Delay function - do nothing for a number of cycles */
while(cycles--);
}
int main(void)
{
status_t error;
/* Configure clocks for PORT */
error = CLOCK_DRV_Init(&clockMan1_InitConfig0);
DEV_ASSERT(error == STATUS_SUCCESS);
/* Set pins as GPIO */
error = PINS_DRV_Init(NUM_OF_CONFIGURED_PINS0, g_pin_mux_InitConfigArr0);
DEV_ASSERT(error == STATUS_SUCCESS);
for (;;)
{
PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 0); // Turn ON green LED (active-low
delay(720000); // Delay while LED is ON
PINS_DRV_WritePin(RGB_LED_GREEN_PORT, RGB_LED_GREEN_PIN, 1); // Turn OFF green LED (active-lo
delay(720000); // Delay while LED is OFF
}
Varia...
Brea...
Expr...
Modu...
Arm ...
Emb...
Expression
Type
Value
Add new expression
Outline
sdk_project_config.h
delay(volatile int) : void
main(void) : int
Dashboard
Project Creation
S32DS Application Project
S32DS Library Project
Build/Debug
Build (All)
Clean (All)
Debug
Settings
Project settings
Build settings
Debug settings
Miscellaneous
Getting Started
Quick access
Console
Registers
Problems
Executables
Debug Shell
Watch registers
Debugger Console
Memory Spaces
hello_world_s32k144 Debug_FLASH [GDB SEGGER J-Link Debugging] JLinkGDBServerCL.exe
Read 2 bytes @ address 0x00000410 (Data = 0xB672)
Setting breakpoint @ address 0x00001498, Size = 2, BPHandle = 0x0001
Starting target CPU...
...Breakpoint reached @ address 0x00001498
Reading all registers
Read 4 bytes @ address 0x00001498 (Data = 0x4810B570)
Removing breakpoint @ address 0x00001498, Size = 2
Starting target CPU...
318M of 1007M

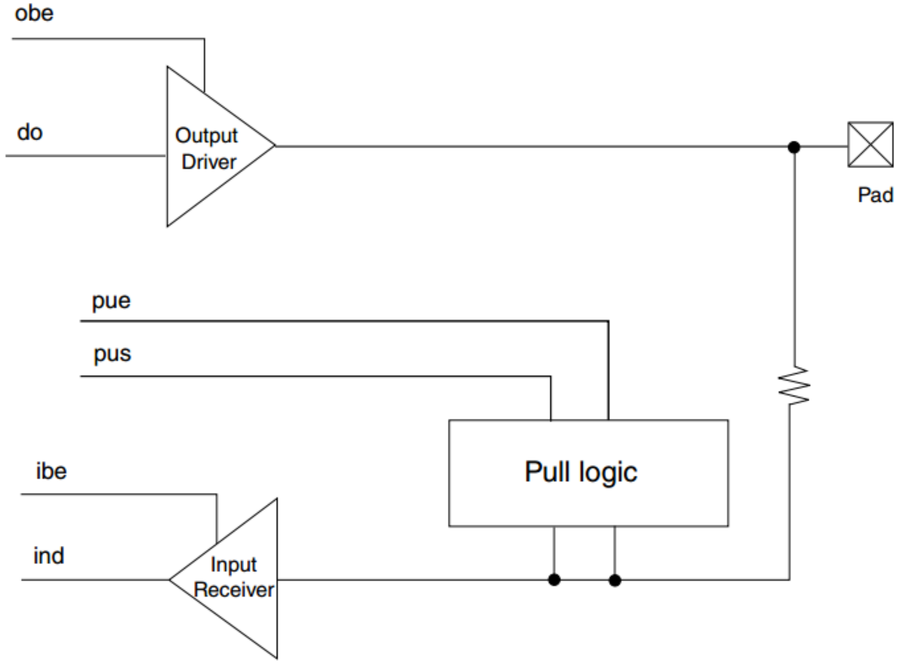
Open C/C++ perspective



Signal Multiplexing



GPIO pad representation



Signal name	Direction	Description
pad	I/O	I/O to external world
do	I	Data coming from the core into the pad
obe	I	Enable output driver
pue	I	0: Disable internal pullup or pulldown resistor 1: Enable internal pullup or pulldown resistor
pus	I	0: Enable internal pulldown resistor if pue is set 1: Enable internal pullup resistor if pue is set
ibe	I	Enable input receiver
ind	O	Data coming out of the pad into the core

Kind Courtesy of reference books/papers/Links:

1. https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-s32-platform:S32DS-S32PLATFORM

Thank you